THE EXACT NONLINEAR RENORMALIZATION STEP FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We prove an exact finite-volume block identity for lattice QED with chamber fermions and an independent external connection. Under explicit domain, primitive and small-charge hypotheses, two finite localizations produce a fluctuation activity whose connected logarithm converges absolutely. We prove its local symmetries, a norm bound after scaling, and a coupling-difference estimate on the unit data ball. The resulting activity, mass, vacuum and external scalar maps reproduce the complete outgoing density and its unnormalized photon-box marginal. The first step integrates all chamber components with their weighted Berezin measure.

Contents

1. Introduction 1
2. Activities, connections and coefficient norms 4
3. The exact block density 8
4. The first chamber step 15
5. Localizing the fluctuation activity 16
6. The connected logarithm 23
7. Symmetries of the local logarithm 27
8. Scaling and coupling estimates 29
9. Identification with the exact density 31
References 32

1. Introduction

We consider one block transformation of noncompact lattice QED with four-component chamber fermions. The photon field is periodic on $\eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, and spinors are antiperiodic in every coordinate. Each photon site carries 48 chamber labels. The chamber measure is $\eta^3/48$; barred and unbarred Grassmann generators are independent. The Dirac stencil, block averages and weighted transposes are those of [FB, §1].

Fix $0 < \rho < 1$, odd $L \geq 4^8$, an integer power M of L with $M \geq L^2$ and $M > 4L$, and $k \in \mathbb{Z}_{\geq 0}$. Put $\eta = L^{-k}$, $e = e_k \in (0,1)$, $e_{k+1} = \sqrt{L}e_k$, and $0 < \mu \leq 1/L$; the scaled reference mass is $L\mu$. Both charges are at most the common ceiling e_{prim} of the photon and fermion estimates specified in the theorem below. Require $\ell/(LM)$ and $\ell/(L^2M)$ to be odd integers at least 13. Fix an integer $r \geq 2$. For $v = -\log e$, put

$$r_k = \min\{L^i : i \geq 0 \text{ an integer},\ L^i \geq v^r\}, \qquad \ell \geq Mr_k.$$

An activity in $K_k^{x,\mathbb{C}}$ is a balanced family of Grassmann polynomials in ordinary and ordered-pair generators, holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, local in its polymer and its connection bonds, and covariant under the lattice, independent local real-gauge and charge-conjugation actions. The real space K_k^x also satisfies the antilinear reality identity. We use exactly the domains, pair support and actions of [NC, §2], recalled in section 2. In particular, $\widetilde{R}_k$ permits independent local real gauges; it imposes no bound on a real harmonic part. With $d_M(X)$ the connected-net length divided by M, the weights

and norm are

$$h_k = (e_k^{-1/4}, e_k^{-1/4+\varepsilon}), \qquad \|\mathscr{E}\|_k^x = \sup_X \exp(\kappa d_M(X)) \sup_{\widetilde{R}_k \times \mathfrak{X}_k} \|\mathscr{E}(X)\|_{h_k}.$$

The coefficient norm is the sum of absolute ordered coefficients times their generator weights. Set

$$w_k^{\mathrm{m}} = e_k^{3/4-8\varepsilon}, \quad w_k^{\mathrm{E}} = e_k^{1/4-7\varepsilon}, \quad \mathrm{Ball}_k^x(c) = \{(m, \mathscr{E}) : |m| \le c w_k^{\mathrm{m}},\ \|\mathscr{E}\|_k^x \le c w_k^{\mathrm{E}}\}. \tag{1}$$

At $k = 0$ the activity is required to be zero. The interaction mass m is distinct from the positive reference mass μ. Put $n_k = \ell^3$, $n_{k+1} = (\ell/L)^3$, and $q = e_k$ at $k = 0$. The real photon domain is $\mathrm{Dom}_j = \{A : H_j^{\mathrm{ph}} A \in \frac{1}{4}\widetilde{R}_j\}$. The density $\mathscr{F}_j^x$ and normalized block operator $\mathbf{T}_k^x$ are defined by (38) and (39); the latter integrates over $\mathrm{Box}_k = \{\sup_\Upsilon |\zeta(\Upsilon)| \le p_{0,k}\}$.

The quantitative constants are

$$\begin{gathered}
\bar{\varepsilon} = 1/200, \qquad 0 < \varepsilon \le \bar{\varepsilon}, \qquad 1 - \varepsilon < \alpha < 1, \\
c_X = 54, \quad \kappa_0 = 54 \log 72, \quad K_0 = \exp(\kappa_0)/36, \quad c_{\mathrm{rb}} = c_X K_0, \quad c_{\mathfrak{fl}} = 61 c_{\mathrm{rb}}, \\
c_C = 3L^{9/2}, \quad b = 1, \quad b_L = b/L, \quad b_- = 1 - L^{-1}, \\
C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)], \\
c_S = 10^5 c_X^5 K_0^5 \exp(6\kappa), \qquad \kappa_w = \kappa + \kappa_0 + 3.
\end{gathered}$$

The primitive constants $C_w, \gamma_w, \alpha_0 > 0$, the photon local-gauge constant C_B, and the two exponential-sum bounds c_γ are those of the photon and fermion families in hypothesis (i) below; $C_w, c_\gamma \ge 1$.

Theorem A (The exact nonlinear step)**.** In the setting above, assume the following simultaneous hypotheses.

(i) Use a single system of photon and fermion primitives on this torus. Assume the full hypotheses and conclusions of [MB, Theorems A–B and Proposition 5.1] and [FB, Theorem A, Propositions 4.1, 4.2, 4.3, 6.1 and 6.2]. Use common constants $(\alpha_0, \gamma_w, \kappa_w; C_w)$ for their analytic kernel and short Hölder bounds on $|s_a| \le M^{\alpha_0}$. This assumption includes component restriction, row vanishing, local read bonds, the exact operators at $s = 1$, and their lattice, gauge, transpose and reality identities. At $k = 0$ retain the stated diagonal families and chamber covariance. Use the weighted integration conventions of [FB, Theorem B and Proposition 5.1] and the extraction, reconstruction and geometry of [NC, Theorems A–B and Appendix B]. The fine inverses required for reconstruction exist at both signs of every total connection in $\widetilde{G}_k$ by the fermion hypothesis.

(ii) The extraction and scaling parameters satisfy

$$\begin{gathered}
L^{1/8-3\bar{\varepsilon}/2} \ge 2400, \qquad e^\varepsilon \le \tfrac{1}{4}, \qquad 160 L^{1+3\varepsilon/2} M e^\varepsilon \le 1, \\
9ML^\varepsilon e^{2\varepsilon} \le 1, \qquad 90 c_X L M e^\varepsilon \le 1, \\
\kappa \ge 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \ge \kappa.
\end{gathered} \tag{2}$$

(iii) The weakening radius and constants satisfy

$$C_w \ge 4\exp(2\gamma_w) M^{27\alpha_0}, \qquad M^{\alpha_0} - 1 \ge \exp\left((2L+1)^3 \kappa + \log(6c_X)\right).$$

For the localizations, connected expansion, scaling and determinant reblocking, assume

$$\begin{gathered}
p_{0,k} \le e^{-\varepsilon/4}, \qquad 12\exp(-p_{0,k}^2/2) \le e^7, \qquad 2e^{5\varepsilon/2} \le 1, \qquad c_{\mathfrak{fl}} e^\varepsilon \le 1, \qquad 3Me^\varepsilon \le 1, \\
2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \le 1, \qquad 16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \le 1, \qquad 96 M^3 e^{1/4} \le 1, \\
32 C_w C_T e^{3\varepsilon/2} \le 1, \qquad 88 M^3 e^{1/4-6\varepsilon} \le 1, \qquad 112 C_w^2 c_\gamma e^{2\varepsilon} \le 3, \\
c_S e^{1/4-5\varepsilon} \le 1, \qquad c_S L^3 e^\varepsilon \le 1, \qquad e^\varepsilon \le \tfrac{1}{4}, \qquad 8 c_{\mathrm{rb}} L^3 C_d M^3 e^{2\varepsilon} \le 1.
\end{gathered} \tag{3}$$

Here C_d is the local determinant constant in hypothesis (i).

(iv) For fixed positive exponents p_0, p, put $p_{0,k} = v^{p_0}$, $p_k = v^p$. The photon box and domain bounds and the determinant gain require

$$p_{0,k} \le e^{-\varepsilon/4}, \qquad 12\exp(-p_{0,k}^2/2) \le e^7, \qquad 8C_B p_k \le e^{-3/4+3\varepsilon},$$
$$8e^{3/4-7\varepsilon/2} \le 1, \qquad L^{3/4} \ge 8/3, \qquad v \ge 9, \qquad 8M(v/4+4)e^{\varepsilon} \le 1,$$
$$e^{1/4} + e^{\varepsilon} < 1, \qquad 4e^{1/4+2\varepsilon} + 2e^{3\varepsilon} \le 1.$$

There is a local connected logarithm $E_k^{\#}(m, \mathscr{E})$ on $\widetilde{R}_k^- \times \mathfrak{X}_k$ for every complex datum in $\mathrm{Ball}_k^x(4)$. It has the lattice symmetry with LM translations, independent local real-gauge covariance and charge conjugation. Each monomial is balanced; real data give the reality identity. Its scaled sum

$$E_k^*(m, \mathscr{E})(Y; A, C) = \sum_{\bar{X}^L = LY} S_L E_k^{\#}(m, \mathscr{E})(X; A_L, C_L)$$

belongs to $K_{k+1}^{x,\mathbb{C}}$ and has norm at most $e_k^{1/4-6\varepsilon}$. Here $A_L(b) = L^{-1/2}A(b/L)$; S_L scales ordinary legs by L^{-1} and pair legs by $L^{-1-\alpha}$, and $\bar{X}^L$ is the union of LM-cubes containing X. The correction is norm continuous on each affine complex line's closed allowed set and norm analytic in its interior. For two data in $\mathrm{Ball}_k^x(1)$,

$$\|E_k^*(m_1, \mathscr{E}_1) - E_k^*(m_2, \mathscr{E}_2)\|_{k+1}^x \le e_k^{1/4-6\varepsilon}\left(\frac{|m_1 - m_2|}{w_k^{\mathrm{m}}} + \frac{\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x}{w_k^{\mathrm{E}}}\right).$$

The second quotient is omitted at $k = 0$.

For the local determinant terms of Proposition 6.1 of [FB], put

$$\delta E_k^{\det}(X; \mathscr{A}, \mathscr{C}) = E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C}), \qquad E_k^{\det}(X, \mathscr{C}) = E_k^d(X, \mathscr{C}) - E_k^d(X, 0).$$

These are scalar activities. We use reblocking $\mathcal{L}$ from (34) of [NC]. The maps R, $(\cdot)^{\mathrm{nat}}$, $m(\cdot)$ and $\varepsilon_{\mathrm{v}}(\cdot)$ are the subtraction, adjustment and extraction maps of [NC, Theorem A]. The outgoing maps are

$$\mathscr{E}^+ = \mathcal{L}(R\mathscr{E})^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m, \mathscr{E}), \tag{4}$$
$$m^+ = L(m + m(\mathscr{E})), \tag{5}$$
$$\varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0), \tag{6}$$
$$\mathfrak{P}_{k+1}(Y, \mathscr{C}') = \sum_{\bar{X}^L = LY} (\mathfrak{P}_k(X, \mathscr{C}'_L) + E_k^{\det}(X, \mathscr{C}'_L)). \tag{7}$$

Here $\mathfrak{P}_k$ is any given family of scalar functions of the external slot.

Let $\varepsilon^{\mathrm{v}}, m$ be real, $\mathscr{E} \in K_k^x$, and $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(1)$, with $\mathscr{E} = 0$ at $k = 0$. For $A' \in \mathrm{Dom}_{k+1}$ with arbitrary harmonic part and $\mathscr{C}' \in \mathfrak{X}_{k+1}$, the exact density satisfies

$$\mathbf{T}_k^x \mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}] = \mathscr{F}_{k+1}^x[\varepsilon^{\mathrm{v},+}, m^+, \mathscr{E}^+], \tag{8}$$

with external scalar families $\mathfrak{P}_k$ and $\mathfrak{P}_{k+1}$ on the respective sides. Its integral satisfies

$$y_{k+1}^x(A'; \mathscr{C}') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta)\, y_k^x(A_k^{\min}(A') + \mathrm{Sq}_k\zeta; \mathscr{C}'_L), \tag{9}$$

where $y_j^x = \int \mathrm{D}\Psi\, \mathscr{F}_j^x$ with the data and scalar family of that level. The photon integral uses the unnormalized Gaussian restriction to the box. Moreover $\mathscr{E}^+ \in K_{k+1}^x$ and

$$\|\mathscr{E}^+\|_{k+1}^x \le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon}\|\mathscr{E}\|_k^x + e_{k+1}^{1/4-2\varepsilon} + e_k^{1/4-6\varepsilon}, \tag{10}$$
$$|m^+ - Lm| \le L c_{\mathfrak{fl}} e_k^{1/2}\|\mathscr{E}\|_k^x, \tag{11}$$
$$|\varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}}| \le L^3(c_{\mathfrak{fl}}\|\mathscr{E}\|_k^x + e_k^7). \tag{12}$$

At level zero this formula integrates all $192n_0$ incoming pairs of generators, with chamber measure $1/48$, and leaves $4n_1$ outgoing pairs. Its data are

$$\mathscr{E}^{+} = \mathcal{L}\delta E_0^{\mathrm{det}} + E_0^{*}(m,0), \quad m^{+} = Lm, \quad \varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \delta\varepsilon_0^0).$$

In particular $0 \le \varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}} \le L^3 q^7$. All maps commute with setting the external connection to zero. The estimates are independent of the side and harmonic part, for one primitive system satisfying the stated hypotheses. The theorem asserts one step on the two admitted sides; it makes no assertion that the outgoing datum lies in the next unit ball. ■□

The proof develops the exact density and nonlinear arguments of [QED, §§13–14]. The quadratic integration in sections 3 and 4 first separates a new fermion density, a determinant quotient and a finite fluctuation integral. The main difficulty is to express that last factor by activities with local coefficients. Two finite localizations in section 5 do this: the first resolves the change of connection and reconstructs the mean field; the second resolves the fermion fluctuations. Component restriction makes disconnected interpolation terms vanish before any infinite expansion is taken.

The resulting activity satisfies

$$\|F(Y)\|_{(h_k/4;\mathsf{w}_W)} \le 36 c_X^2 K_0^2 e_k^{1/4-5\varepsilon} \exp[-(\kappa - 2\kappa_0 - 2) d_M(Y)].$$

The connected-polymer expansion in section 6 converges absolutely, with a geometric tail in the number of polymers. Its exponential equals the original finite integral by absolute regrouping. For charge conjugation, the unbarred fluctuation contains Γ_k whereas the barred one does not. Equality of their transformed Wick covariances, followed by inversion of a finite subset sum, gives the symmetry of each local logarithm. Scaling and Cauchy's formula give the correction and coupling bounds. Section 9 evaluates the two finite localization sums and recovers the density identity with its determinant and photon-box factors.

The extraction and reconstruction arguments are proved separately in [NC]; their exact hypotheses are retained above. The elementary labelled-tree degree formula used in the connected sum is the Prüfer formula in [Pos19, Proposition 4]. All later sections use the hypotheses of theorem A. No limiting trajectory is part of the present theorem.

2. Activities, connections and coefficient norms

We specify the local activity space used in theorem A. These are the conventions of [NC, §2].

2.1. **Chamber sites, pairs and connection domains.** Write O_{cub} for the signed permutation matrices on $\mathbb{R}^3$. A chamber site is $x = (c, g)$, with photon projection $\pi x = c$ and $g \in O_{\mathrm{cub}}$. Integration over chamber sites uses $\mathrm{d}x = (\eta_k^3/48)\sum_{c,g}$. We take M-cubes centered at $M\mathbb{Z}^3/\ell\mathbb{Z}^3$; their lattice sites form a partition. Blocks of every odd side used below are centered in the same way.

For the three internal neighbors let s_0 change the sign of the third coordinate, let s_1 exchange the second and third coordinates, and let s_2 exchange the first and second. Let r_{e_1} change the sign of the first coordinate. Set

$$x^{(i)} = (c, gs_i) \quad (0 \le i \le 2), \qquad x^{(3)} = (c + \eta_k g e_1, g r_{e_1}).$$

Each neighbor map is an involution. An oriented bond field B is antisymmetric under reversal and periodic on the photon lattice. Write $B_\mu(c) = B(c, c + \eta_k e_\mu)$, and use the forward differences

$$\partial_\nu B_\mu(c) = \eta_k^{-1}(B_\mu(c + \eta_k e_\nu) - B_\mu(c)), \qquad (d\lambda)_\mu(c) = \eta_k^{-1}(\lambda(c + \eta_k e_\mu) - \lambda(c)).$$

Thus $(dB)_{\nu\mu} = \partial_\nu B_\mu - \partial_\mu B_\nu$. Distances between photon sites are sup distances, using the shortest lift when a distance bound is imposed. For $0 < \alpha < 1$ the three norms are

$$\|B\|_1 = \|B\|_\infty, \qquad \|B\|_2 = \|\partial B\|_\infty, \qquad \|B\|_3 = \|\delta_\alpha \partial B\|_\infty = \sup_{\substack{\mu,\nu,c,c' \\ 0<d(c,c')\le 1}} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c,c')^\alpha}.$$

Local versions use only the indicated cells in the given region. Let $\square^\dagger$ be the concentric enlargement of an M-cube to side $(2L+1)M$. Put $\theta_i = e_k^{-3/4+i\varepsilon}$. The connection domains are

$$\begin{aligned} R_k &= \{B : \|B\|_i < \theta_i,\ 1 \le i \le 3\}, \\ \widetilde{R}_k &= \{B_0 + B_1 : B_1 \in R_k,\ B_0 \text{ real, and on each } \square^\dagger, \\ &\qquad\qquad B_0 = B^{\mathrm{b}}_\square + d\lambda_\square \text{ with } B^{\mathrm{b}}_\square, \lambda_\square \text{ real and } \|B^{\mathrm{b}}_\square\|_i < \theta_i\}, \\ \mathfrak{X}_k &= \{C : \|C\|_\infty < e_k^{-1+2\varepsilon},\ \|\partial C\|_\infty < e_k^{-1+3\varepsilon}\}. \end{aligned} \tag{13}$$

Only existence of these representatives is required. For $0 < c \le 1$, the notation $c\widetilde{R}_k$ multiplies all three bounds for both summands by c. For $\widetilde{R}_k^-$ the local real bounds remain θ_i, while the global summand has bounds $(1 - e_k^\varepsilon)\theta_i$.

For a short ordered pair $x = (c, g), x' = (c', g')$, let $\tau B(c, c')$ be the arithmetic mean of the sums of B along the six coordinate-order paths from c to c'. The bond values in this sum are unscaled; define

$$U_B(x, x') = \exp\{ie_k \eta_k \tau B(c, c')\}, \qquad (\delta_{\alpha,B}\psi)(x, x') = \frac{U_B(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta_k)^\alpha}.$$

For pairs within one cluster the transport is one. Barred fields use U_B^{-1}. These formulas specify evaluation; the pair generators in the algebra remain independent.

We use Pauli matrices and

$$\gamma_\mu = \sigma_3 \otimes \sigma_\mu, \quad \gamma_5 = \sigma_1 \otimes 1_2, \quad \Sigma_3 = \sigma_3 \otimes 1_2, \qquad \mathsf{C} = i\gamma_2, \quad \varkappa = \gamma_5 \mathsf{C} = \sigma_2 \otimes \sigma_2.$$

Here C is real with $\mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}$, while $\varkappa$ is a real symmetric involution. The spin group below is the group generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_5$. Its diagonal coordinate action $r(S)$ is determined by $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Connections transform by pullback on oriented bonds.

2.2. **The local Grassmann algebra.** Fix a level k, mesh $\eta_k = L^{-k}$ and an admissible side ℓ, with $\ell/(LM)$ odd and at least 13. Let $\mathcal{P}_k$ be the collection of face-connected polymers of M-cubes on this torus. A polymer is small when $d_M(X) \le L$. A small polymer has an injective lift, unique up to a period translation, by the diameter and lifting estimates of Appendix B of [NC]. Write $\mathrm{Vol}(X) = M^3|X|_M$.

The independent Grassmann generators are ψ, $\bar\psi$ on the chamber sites $x = (c, g)$ and χ, $\bar\chi$ on ordered pairs. For distinct chamber sites on the covering lattice, put $d_X(x, x') = \max(|c - c'|_\infty, \eta_k)$ and restrict pairs to $d_X \le M$. Pairs on the torus are simultaneous period classes; their shortest lifts are unique since $2M < \ell$. A pair is admissible for X when its photon sites and all six rectilinear paths between them lie in X, or when it is $(x, x^{(3)})$ and one of its photon sites belongs to X. Thus a pair may leave X only along a bond of type 3. Pairs inside a cluster have $d_X = \eta_k$ and are admissible when their common photon site belongs to X. Ordinary generators acquire the sign $(-1)^{n_1+n_2+n_3}$ under translation by ℓn. Pair generators acquire the same sign when both endpoints are translated by ℓn. The kernels are antiperiodic in each ordinary argument and in each simultaneous pair argument with these conventions.

Let $\xi = (x, \beta, \omega)$ index an ordinary leg and $\zeta = (x, x', \beta, \omega)$ a Hölder leg, where $\omega = 0$ is unbarred and $\omega = 1$ is barred. We use the index measures

$$\nu(\xi) = \eta_k^3/48, \qquad \nu(\zeta) = (\eta_k^3/48)^2.$$

Integrals sum the spin and bar indices as well. An activity has a unique expansion with separately antisymmetric kernels,

$$\mathscr{E}(X;\mathscr{A},\mathscr{C}) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, \mathscr{E}_{nm}(X;\mathscr{A},\mathscr{C};\xi;\zeta)\psi(\xi_1)\cdots\psi(\xi_n)\chi(\zeta_1)\cdots\chi(\zeta_m). \tag{14}$$

Barred generators are included in this notation. The generator weights, as distinct from the index measures, are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}.$$

For an element F the weighted coefficient norm is

$$\|F\|_{h_k} = \sum_{n,m} \frac{h_{k,1}^n h_{k,2}^m}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, |F_{nm}|. \tag{15}$$

In particular a generator has norm $h_{k,i}$; the weight ν enters only when coefficients are written as integral kernels.

An activity family in $K_k^{x,\mathbb{C}}$ is balanced: every monomial has as many barred as unbarred legs. Its kernels are holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, vanish unless all ordinary legs lie over X and all pairs are admissible for X, and read the two connections only on bonds meeting X. The norm is

$$\|\mathscr{E}\|_k^x = \sup_{X\in\mathcal{P}_k} \exp(\kappa d_M(X)) \sup_{(\mathscr{A},\mathscr{C})\in\widetilde{R}_k\times\mathfrak{X}_k} \|\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k}. \tag{16}$$

We require $\kappa \geq \kappa_0$ and finite norm.

The required symmetries act on both the connections and the generators. For $\tilde{g} = (S, a)$ with the coordinate sign action $r(S)$ and $a \in M\mathbb{Z}^3$, put

$$\tilde{g}(c,g) = (r(S)c + a, r(S)g), \qquad a_{\tilde{g}}\psi(x) = S\psi(\tilde{g}^{-1}x), \qquad a_{\tilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\tilde{g}^{-1}x).$$

The same spin matrices act on pair generators after transforming both endpoints. Local gauge transformations multiply an unbarred ordinary or pair leg by $\exp(ie_k\Lambda(c))$, using the photon site of its first point, and a barred leg by the inverse phase. Write this automorphism as G_Λ. Charge conjugation and the antilinear reality map act by

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \qquad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \qquad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

and by the same rules on the Hölder generators.

The covariance requirements on a family are

$$\begin{aligned} a_{\tilde{g}}\mathscr{E}(\tilde{g}X;\tilde{g}\mathscr{A},\tilde{g}\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}), \\ \mathrm{Cg}\,\mathscr{E}(X;-\mathscr{A},-\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}). \end{aligned} \tag{17}$$

For real scalars λ, λ' and two pairs in the domain which satisfy $\mathscr{A}^1 = \mathscr{A} - d\lambda$ and $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on bonds at distance at most one from X, the local gauge requirement is

$$G_{\lambda+\lambda'}\mathscr{E}(X;\mathscr{A}^1,\mathscr{C}^1) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

The real subspace K_k^x has in addition

$$\mathbf{K}\mathscr{E}(X;-\overline{\mathscr{A}},-\overline{\mathscr{C}}) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

All requirements refer to the generators before evaluating fields. They are essential to the form of the counterterms.

The spaces $K_k^{\mathbb{C}}, K_k$ without the second connection argument have the same definitions on $\widetilde{R}_k$. Restriction $\mathscr{E}|_0(X;\mathscr{A}) = \mathscr{E}(X;\mathscr{A},0)$ maps $K_k^{x,\mathbb{C}}$ into $K_k^{\mathbb{C}}$ and has norm at most one. The real subspaces also restrict. Analyticity, locality and the symmetries follow by setting the second argument and its gauge function to zero; the norm bound follows from the smaller supremum.

2.3. **Total connections.** The comparison domain $\widetilde{G}_k$ consists of fields $B_0 + B_1$ with B_0 real and $\|B_1\|_\infty < e_k^{-1+\varepsilon}$, where on each $\square^{\sim 5}$ the field B_0 has a real-gauge representative of supremum norm below $e_k^{-1+\varepsilon}$. Here $\square^{\sim 5}$ has the same center and side $11M$. The real-gauge representatives in the domains defined above may be chosen independently on each enlarged cube.

Assume $L \geq 5$, $0 < \varepsilon \leq 1/200$ and

$$e_k^{\varepsilon} \leq \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \leq 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \leq 1. \tag{18}$$

Then

$$\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k. \tag{19}$$

Indeed, write $\mathscr{A} = A_0 + A_1$ with $A_1 \in R_k$ and A_0 locally real-gauge equivalent on $\square^\dagger$ to $A^b_\square$ with the three threshold bounds. Restrict the representative to $\square^{\sim 5} \subset \square^\dagger$. Its supremum norm is below $e_k^{-3/4+\varepsilon} \leq e_k^{-1+\varepsilon}$, and

$$\|A_1 + \mathscr{C}\|_\infty < e_k^{-3/4+\varepsilon} + e_k^{-1+2\varepsilon} = (e_k^{1/4} + e_k^{\varepsilon}) e_k^{-1+\varepsilon} \leq e_k^{-1+\varepsilon}.$$

This is the required decomposition of the total connection. Real gauge transformations can absorb any real constant part, locally on the nonwrapping cubes.

Put $\mathfrak{i}(B) = e_k \| \operatorname{Im} B\|_\infty$ and $q_d(B) = e_k \|dB\|_\infty$. The same decomposition gives

$$\mathfrak{i}(\mathscr{A} + \mathscr{C}) \leq e_k^{1/4+\varepsilon} + e_k^{2\varepsilon} \leq 1, \qquad q_d(\mathscr{A} + \mathscr{C}) \leq 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \leq 1. \tag{20}$$

For the curvature, every plaquette lies in an enlarged cube, and $dA_0 = dA^b_\square$ there. This costs at most $2\|\partial A^b_\square\|_\infty$; the global small field costs another $2\|\partial A_1\|_\infty$, and the external field costs $2\|\partial \mathscr{C}\|_\infty$. The imaginary bound uses only that A_0 is real.

The domains are open: the strict inequalities define open sets for the global summand, which can absorb a sufficiently small complex perturbation while the local real representative is kept fixed. The norm-defined sets are invariant under sign and conjugation; the local real-gauge definitions are also invariant under real gauges and real constant additions. Coordinate sign changes and M translations preserve them. Their derivative and Hölder norms are preserved, with a possible reversal of a bond and translation of its base point. Thus (19) applies also at the signs, conjugates and real gauges used below.

2.4. **The operator and photon data.** The injection and cluster mean are $(Jv)(c,g) = v(c)$ and $(J^* f)(c) = 48^{-1} \sum_g f(c,g)$. The normalized average $\mathcal{Q}_k = Q_k J^*$, its reverse $\mathcal{Q}_k^\sharp = \mathcal{Q}_k(-B)^T$, the one-step average Q, and the chamber operator D_B are given in [FB, §1]. Put $b_k = (1-L^{-1})/(1-L^{-k})$ for $k \geq 1$. At these levels set

$$S_k(B) = (D_B + \mu + b_k \mathcal{Q}_k^\sharp \mathcal{Q}_k)^{-1}, \quad H_k(B) = b_k S_k(B) \mathcal{Q}_k^\sharp(B), \quad D_k(B) = b_k - b_k^2 \mathcal{Q}_k(B) S_k(B) \mathcal{Q}_k^\sharp(B).$$

The one-step projection is $P_L(B) = Q^T(-B)Q(B)$. The barred map is $\bar{H}_k(B) = \mathsf{C}^{-1} H_k(-B) \mathsf{C}$. At $k = 0$, $H_0 = \bar{H}_0 = 1$ and $D_0(B) = D_B + \mu$. The symbol $\mathcal{S}(a, \ell)$ denotes the four-component spinor space on the a-spaced torus, with measure a^3; the superscript ch adds chamber labels and divides the measure by 48. All spaces have the antiperiodic convention stated above. For later use, the hopping notation of [FB, §2.1] is

$$R_i(B) f(x) = U_B(x, x^{(i)}) f(x^{(i)}), \qquad \mathcal{L}_B = \sum_{i=0}^{3} w_i (1 - R_i(B)), \qquad M_i(c,g) = \tfrac{w_i}{2}(\rho + \gamma \cdot g\mathbf{n}_i),$$

where $(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6)$ and $(\mathbf{n}_0, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1)$. Thus $\widehat{d}_i(c,g) = -g\mathbf{n}_i$. On a nearest-neighbor photon bond write $u_B(c,c') = \exp(i e_k \eta_k B(c,c'))$; this is the transport $U_B(x, x^{(3)})$ when c, c' are its endpoints.

For photon fields use the Landau reconstruction H_k^{ph}, the constrained minimum $A_k^{\min}$ and the real coordinate map $C : V_k \to \mathcal{Z}$ of [MB, Theorem A]. Write $C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}$ and $\mathrm{Sq}_k = C C_k^{1/2}$. The

coordinate basis $\{\Upsilon\}$ is orthonormal and has the specified block or face-bond supports. The law μ_I is the product standard Gaussian in these coordinates. Its restriction to Box_k has probability $\exp(-\ell^3\delta\varepsilon_k^0)$, with $0 \le \delta\varepsilon_k^0 \le 12\exp(-p_{0,k}^2/2)$. The symbols C and $\mathscr{C}$ always denote, respectively, this real coordinate map and the external connection.

The common constant c_γ bounds both

$$\sup_{\Upsilon}\sum_{\Upsilon'}\exp[-\gamma_w d(\Upsilon,\Upsilon')], \qquad \sup_{b}\sum_{b'}\exp[-\gamma_w d(b,b')],$$

where the second sum is over unit bonds. The constants C_w and γ_w dominate the photon bounds and the corresponding fermion bounds as stipulated in hypothesis (i).

3. The exact block density

The block transformation has three factors: a new quadratic fermion density, a determinant quotient, and a finite integral over photon and fermion fluctuations. We derive this identity before expanding its last factor. The integration at level zero includes every chamber component and uses the chamber Berezin measure. The algebraic identities in this section follow [QED, §13.1–13.3].

3.1. **Integration spaces and real backgrounds.** Fix $k \in \mathbb{Z}_{\ge 0}$, $\eta_k = L^{-k}$, odd $L \ge 5$, and an integer power M of L with $M \ge L^2$, with $\ell/(LM)$ odd and at least 13. The regulator parameter satisfies $0 < \rho < 1$. Keep $b = 1$ as a letter, and write $b_L = b/L$ and $b_k = b(1-L^{-1})/(1-L^{-k})$ for $k \ge 1$. Write $\mathscr{V}_k = \mathcal{S}(1,\ell)$ for $k \ge 1$, with four spin components and counting measure, and $\mathscr{V}_0 = \mathcal{S}^{\mathrm{ch}}(1,\ell)$, with 192 components per photon site and measure $1/48$. The one-step averaging operator into $\mathcal{S}(L,\ell)$ is

$$\mathcal{A}_k(B) = \begin{cases} Q(B), & k \ge 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{A}_k^\sharp(B) = \mathcal{A}_k(-B)^T.$$

The target measure is L^3. On these spaces set

$$\mathcal{D}_k(B) = \begin{cases} D_k(B), & k \ge 1, \\ D_B + \mu, & k = 0, \end{cases} \qquad K_k(B) = \mathcal{D}_k(B) + b_L\mathcal{A}_k^\sharp(B)\mathcal{A}_k(B). \tag{21}$$

Thus $K_k = D_k + b_L P_L$ for $k \ge 1$ and $K_0 = F_0 = D_B + \mu + b_L J P_L J^*$. We write $D_0(B) = D_B + \mu$, so that $\mathcal{D}_k(B) = D_k(B)$ at every level. When it exists, $\Gamma_k = K_k^{-1}$, and $\delta Z_k(B) = \det K_k(B)$. The transpose refers to the displayed bilinear measures. For a real B, the phases have modulus one and $\mathcal{A}_k^\sharp = \mathcal{A}_k^*$.

The normalized averages give $Q(B)Q^T(-B) = 1$. On a coarse block $B(z')$ the product is $L^{-3}\sum_{z\in B(z')} U_B(z',z)U_B(z',z)^{-1} = 1$. At level zero the additional cluster mean contributes $48/48 = 1$. Similarly $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$ for $k \ge 1$. In a unit block the photon count gives $\eta_k^3\eta_k^{-3} = 1$ and the cluster count again gives $48/48 = 1$. For real B the inverse phases in the embeddings are the complex conjugates of the averaging phases. The products $\Pi_1 = \mathcal{Q}_k^\sharp\mathcal{Q}_k$ and $\Pi_L = \mathcal{Q}_k^\sharp P_L\mathcal{Q}_k$ are therefore orthogonal projections; so are P_L and, at level zero, JP_LJ^*. Their scalar spin kernels commute with every constant spin matrix.

The fine resolvents used below are those of [FB, §4.2]. Explicitly, for $k \ge 1$ put

$$F_\infty(B) = D_B + \mu + b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k, \quad \mathcal{B}_{k,y} = (b_k + i\gamma_5 y)^{-1}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-1}P_L,$$

and define

$$F_y = F_\infty - b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_{k,y}\mathcal{Q}_k, \qquad S_{k,y} = F_y^{-1}$$

where the inverse exists. In particular $S_k = F_\infty^{-1}$. The reconstruction maps are $T_k(B) = \mathcal{Q}_k(B)(b_k^{-1}D_B + 1 + \mu/b_k)$ and $\bar{T}_k(B) = \mathsf{C}^{-1}T_k(-B)\mathsf{C}$; their recovery identities are [NC, Theorem B(iv)]. At zero these reconstruction maps are identities on chamber fields.

Proposition 3.1. Let B be real and $\mu > 0$. The operators $D_B + \mu$, $\mathcal{D}_k(B)$ and $K_k(B)$ are invertible at every level at which they are defined. For $k \geq 1$,

$$D_k(B)^{-1} = b_k^{-1} + \mathcal{Q}_k(B)(D_B + \mu)^{-1}\mathcal{Q}_k^\sharp(B), \qquad \operatorname{Re} D_k(B)^{-1} \geq b_k^{-1}. \tag{22}$$

Moreover $\operatorname{Re} D_k > 0$ and $\operatorname{Re} K_k > 0$ as quadratic forms for $k \geq 1$, whereas $\operatorname{Re} K_0 \geq \mu$. Each K_k satisfies $K_k^* = \gamma_5 K_k \gamma_5$. For every real y, $K_k + i\gamma_5 y$ is invertible, and for $y \neq 0$,

$$\|(K_k + i\gamma_5 y)^{-1}\|_{2\to 2} \leq |y|^{-1}. \tag{23}$$

For $k \geq 1$ the fine inverse $S_{k,y}$ also exists for every real y. In particular $\det D_k(0)$ and $\delta Z_k(0)$ are nonzero, including the determinants on $\mathscr{V}_0$. ■

Proof. At a real connection the chamber hops are self-adjoint unitary involutions. In the proof of (6) of [FB], reindexing x by $x^{(i)}$ uses $U_B(x^{(i)}, x) = \overline{U_B(x, x^{(i)})}$. It gives

$$(f, \mathcal{L}_B f) = \frac{1}{2}\sum_i w_i \int \mathrm{d}x\, |R_i(B)f(x) - f(x)|_2^2 \geq 0.$$

The direction vector changes sign under the same involution; hence $\sum_i w_i(\gamma \cdot \widehat{d}_i)R_i(B)$ is anti-self-adjoint. The cancellation $\sum_i w_i \widehat{d}_i = 0$ yields $D_B^* = \gamma_5 D_B \gamma_5$ and $\operatorname{Re} D_B = (\rho/(2\eta_k))\mathcal{L}_B \geq 0$. Thus $\operatorname{Re}(D_B + \mu) \geq \mu$. For $k \geq 1$ adding the positive penalty $b_k\Pi_1$ gives $\operatorname{Re} F_\infty \geq \mu$. These inequalities imply injectivity and then invertibility in the finite-dimensional spaces.

Put $G = D_B + \mu$ and $S_k = (G + b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k)^{-1}$. Multiplication of $b_k^{-1} + \mathcal{Q}_k G^{-1}\mathcal{Q}_k^\sharp$ by $D_k = b_k - b_k^2\mathcal{Q}_k S_k \mathcal{Q}_k^\sharp$ gives

$$1 + b_k\mathcal{Q}_k G^{-1}[1 - (G + b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k)S_k]\mathcal{Q}_k^\sharp = 1.$$

This proves (22). For an invertible operator O, $\operatorname{Re}(u, Ou) = \operatorname{Re}(Ou, O^{-1}Ou)$. In particular $\operatorname{Re} O \geq 0$ implies $\operatorname{Re} O^{-1} \geq 0$. As $\mathcal{Q}_k^\sharp = \mathcal{Q}_k^*$ here, the second term of D_k^{-1} has nonnegative real part. For a nonzero v, consequently,

$$\operatorname{Re}(v, D_k v) = \operatorname{Re}(D_k v, D_k^{-1} D_k v) \geq b_k^{-1}\|D_k v\|_2^2 > 0.$$

Adding $b_L P_L$ preserves strict positivity. At $k = 0$ the same conclusion follows directly by adding $b_L J P_L J^*$ to $D_B + \mu$.

The projections commute with γ_5. It follows first that $F_\infty^* = \gamma_5 F_\infty \gamma_5$ and $S_k^* = \gamma_5 S_k \gamma_5$. Taking the adjoint in the formula for D_k and using $\mathcal{Q}_k^\sharp = \mathcal{Q}_k^*$ gives $D_k^* = \gamma_5 D_k \gamma_5$. The penalty has the same symmetry, proving it for K_k. For $k = 0$ this is already the symmetry of its three terms. Hence

$$(K_k + i\gamma_5 y)^*(K_k + i\gamma_5 y) = K_k^* K_k + y^2, \qquad \|(K_k + i\gamma_5 y)f\|_2^2 = \|K_k f\|_2^2 + y^2\|f\|_2^2.$$

For $y \neq 0$ this proves injectivity and the inverse bound; $y = 0$ is covered by strict positivity. For the fine inverse at $k \geq 1$, use the block matrix with diagonal blocks $b_k + b_L P_L + i\gamma_5 y$ and F_∞, and off-diagonal blocks $-b_k\mathcal{Q}_k$ and $-b_k\mathcal{Q}_k^\sharp$. The first diagonal block is invertible by its idempotent decomposition (53) of [FB]. Its two Schur complements are F_y and $K_k + i\gamma_5 y$. The latter is invertible, so the block matrix and then F_y are invertible. This proves existence of $S_{k,y}$. The asserted normalizing determinants are therefore nonzero at the real zero connection. □

3.2. **The next quadratic density.** Assume now that ℓ/L is also admissible, and let $B = B'_L$. At the outgoing level the mass is $L\mu$ and the charge is $e_{k+1} = L^{1/2}e_k$. The following identities hold whenever $S_k(B)$ and $\Gamma_k(B)$ exist for $k \geq 1$, or $\Gamma_0(B)$ exists for $k = 0$. This invertibility domain is open and invariant under $B \mapsto -B$: all elementary entries are entire functions of the bond values, and the transpose identities are

$$\mathcal{D}_k(B)^T = \mathsf{C}^{-1}\mathcal{D}_k(-B)\mathsf{C}, \qquad K_k(B)^T = \mathsf{C}^{-1}K_k(-B)\mathsf{C}, \qquad \Gamma_k(B)^T = \mathsf{C}^{-1}\Gamma_k(-B)\mathsf{C}. \tag{24}$$

For the first identity reindex each hopping term by $x^{(i)}$, using $M_i(x^{(i)})^T = \mathsf{C}^{-1}M_i(x)\mathsf{C}$ and the reverse phase at $-B$. Transposition of the averages reverses their connection in the product P_L,

and the scalar kernels commute with C. The remaining identities follow by products and inversion. Invertible matrices have analytic inverses on their open determinant domains.

Define the fine-to-L average and its embedding by

$$\mathcal{Q}_k^+(B) = \begin{cases} Q(B)\mathcal{Q}_k(B), & k \geq 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{Q}_k^{+,\sharp}(B) = \mathcal{Q}_k^+(-B)^T.$$

They obey $\mathcal{Q}_k^+\mathcal{Q}_k^{+,\sharp} = 1$. Set $\bar{b}_k = b_{k+1}/L$ and

$$F_k^+(B) = D_B + \mu + \bar{b}_k\mathcal{Q}_k^{+,\sharp}\mathcal{Q}_k^+, \qquad \Sigma_k^+(B) = F_k^+(B)^{-1}. \tag{25}$$

For $k = 0$ this is F_0 and its inverse Γ_0. For $k \geq 1$ it is the fine resolvent operator at $y = 0$, since $\bar{b}_k = b_k b_L/(b_k + b_L)$.

To verify existence and the identities needed below, write $\mathcal{B}_k = (b_k + b_L P_L)^{-1}$ for $k \geq 1$ and consider

$$\begin{pmatrix} b_k + b_L P_L & -b_k\mathcal{Q}_k \\ -b_k\mathcal{Q}_k^\sharp & F_\infty \end{pmatrix}.$$

Its lower-right Schur complement is K_k. Since P_L is idempotent and scalar in spin, $\mathcal{B}_k = b_k^{-1}(1 - P_L) + (b_k + b_L)^{-1}P_L$. The other Schur complement is

$$F_\infty - b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_k\mathcal{Q}_k = D_B + \mu + \frac{b_k b_L}{b_k + b_L}\mathcal{Q}_k^\sharp P_L\mathcal{Q}_k = F_k^+.$$

The two assumed inverses make the block matrix invertible, and hence F_k^+ is invertible. Equating the upper-left and lower-left blocks of its inverse gives

$$\Gamma_k = \mathcal{B}_k + b_k^2\mathcal{B}_k\mathcal{Q}_k\Sigma_k^+\mathcal{Q}_k^\sharp\mathcal{B}_k, \qquad H_k\Gamma_k = b_k\Sigma_k^+\mathcal{Q}_k^\sharp\mathcal{B}_k. \tag{26}$$

This is the same Schur calculation as Proposition 4.2 of [FB], retaining the off-diagonal block for the mean field.

Let

$$D_{k+1}^0(B) = b_L - b_L^2\mathcal{A}_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B). \tag{27}$$

For $k \geq 1$, $QP_L = Q$ and $P_L Q^T(-B) = Q^T(-B)$ give $Q\mathcal{B}_k = (b_k + b_L)^{-1}Q$ and $\mathcal{B}_k Q^T(-B) = (b_k + b_L)^{-1}Q^T(-B)$. Insert these in (26) to obtain

$$\begin{aligned} D_{k+1}^0(B) &= \bar{b}_k - \bar{b}_k^2\mathcal{Q}_k^+\Sigma_k^+\mathcal{Q}_k^{+,\sharp}, \\ b_L H_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B) &= \bar{b}_k\Sigma_k^+\mathcal{Q}_k^{+,\sharp}. \end{aligned} \tag{28}$$

For $k = 0$ both formulas follow directly from $\bar{b}_0 = b_L$, $H_0 = 1$, $\Sigma_0^+ = \Gamma_0$, and $\mathcal{Q}_0^+ = \mathcal{A}_0$.

Let R_L denote the scaling $f_L(c, g) = L^{-1}f(c/L, g)$ on chamber spinors and $v_L(z') = L^{-1}v(z'/L)$ on unit spinors. These are bijections between the corresponding incoming and outgoing spaces. Scaling preserves the label and the nested block centers, and $e_k\eta_k L^{-1/2} = e_{k+1}\eta_{k+1}$ preserves each transport phase. Consequently

$$(D_B + \mu)\mathsf{R}_L = L^{-1}\mathsf{R}_L(D_{B'} + L\mu), \qquad \mathcal{Q}_k^+(B)\mathsf{R}_L = \mathsf{R}_L\mathcal{Q}_{k+1}(B'),$$

and the corresponding embedding identity holds. For the average, its L^{-3} factor cancels the L^3 change of the chamber index measure. Thus

$$F_k^+(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L F_\infty(B'), \qquad \Sigma_k^+(B)\mathsf{R}_L = L\mathsf{R}_L S_{k+1}(B'). \tag{29}$$

In particular the outgoing inverse exists. Equations (28) and (29) now imply

$$D_{k+1}^0(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L D_{k+1}(B'), \qquad b_L H_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B)\mathsf{R}_L = \mathsf{R}_L H_{k+1}(B'). \tag{30}$$

All expressions in the second identity act on the outgoing four-component field.

For independent outgoing Grassmann fields define

$$\Psi^c = b_L \Gamma_k(B) \mathcal{A}_k^\sharp(B) \Psi'_L, \qquad \bar\Psi^c = b_L \Gamma_k(B)^T \mathcal{A}_k(B)^T \bar\Psi'_L. \tag{31}$$

Then, with $\psi' = H_{k+1}(B')\Psi'$ and $\bar\psi' = \bar H_{k+1}(B')\bar\Psi'$,

$$H_k(B)\Psi^c = \psi'_L, \qquad \bar H_k(B)\bar\Psi^c = \bar\psi'_L. \tag{32}$$

The first relation is (30). For the second use (24) and apply the first relation at $-B$ to $\mathsf{C}\bar\Psi'$; the constant matrix C commutes with scaling and the averages. The covariant Hölder quotients of these induced fields are also the scaled outgoing quotients: the transports agree and d_X is divided by L, so the quotient carries the factor $L^{-1-\alpha}$. This holds for every pair, including a pair inside one cluster. Applying $T_k H_k = 1$ and its barred version shows that the means in (31) are also $T_k(B)\psi'_L$ and $\bar T_k(B)\bar\psi'_L$. At level zero these maps are identities on the chamber field.

The quadratic term in the Gaussian block formula is therefore exactly

$$\langle \bar\Psi'_L, D^0_{k+1}(B)\Psi'_L\rangle_L = \langle \bar\Psi', D_{k+1}(B')\Psi'\rangle_1,$$

since the pairings of scaled fields multiply by L. Together with (73) of [FB], this identifies the entire outgoing quadratic density.

Proposition 3.2. On the invertibility domain just specified,

$$\det D_{k+1}(B') = b^{4n_{k+1}} \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}, \qquad n_k = \ell^3, \quad n_{k+1} = (\ell/L)^3. \tag{33}$$

In particular, with $c_k = b^{-4n_{k+1}}$,

$$\frac{c_k}{\det D_k(0)} = \frac{1}{\det D_{k+1}(0)\delta Z_k(0)}. \tag{34}$$

■

Proof. Apply Sylvester's determinant identity to $b_L \mathcal{A}_k \Gamma_k$ and $\mathcal{A}_k^\sharp$ on the two different spaces. It gives

$$b_L^{-4n_{k+1}} \det D^0_{k+1}(B) = \det(1 - b_L \mathcal{A}_k^\sharp \mathcal{A}_k \Gamma_k) = \det(\mathcal{D}_k \Gamma_k) = \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}.$$

The incoming determinant at $k = 0$ has size $192 n_0$; the exponent of b_L is the outgoing dimension $4n_1$, as in every other step. Equation (30) gives $\det D^0_{k+1}(B) = L^{-4n_{k+1}} \det D_{k+1}(B')$. Using $L b_L = b$ proves (33). At the zero connection all denominators are nonzero by proposition 3.1, so (34) follows. □

3.3. **The density with an external connection.** For this subsection assume $0 < \mu \le 1/L$ and $e_{k+1} = L^{1/2} e_k \le e_{\text{prim}}$, with the hypotheses of Propositions 3.3 of [FB], 4.2 of [FB] and Theorem B of [MB]. We also impose (43) of [MB] and (18). The inverse domain $\widetilde G_k$ is the domain $\widetilde{\mathcal{G}}_k$ of the propagator estimates. Let $A' \in \mathrm{Dom}_{k+1}$ be a real block field, with arbitrary harmonic part, and let $\mathscr{C}' \in \mathfrak{X}_{k+1}$. Put

$$\mathscr{A}' = H^{\text{ph}}_{k+1} A', \qquad \mathscr{A} = \mathscr{A}'_L, \qquad \mathscr{C} = \mathscr{C}'_L, \qquad \mathscr{T}' = \mathscr{A}' + \mathscr{C}', \qquad \mathscr{T} = \mathscr{T}'_L.$$

The photon block objects are

$$\begin{aligned} A_k(\zeta) &= A_k^{\min}(A') + \mathrm{Sq}_k \zeta, \\ \mathscr{Z}(\zeta) &= H_k^{\text{ph}} \mathrm{Sq}_k \zeta, \qquad \mathscr{B}_k = H_k^{\text{ph}} A_k^{\min}(A') = \mathscr{A} - d\omega_k. \end{aligned} \tag{35}$$

Here ω_k is real and periodic, as in (30) of [MB]. The truncated photon measure has the normalization

$$\mu_I(\mathrm{Box}_k) = \exp(-n_k \delta\varepsilon_k^0), \qquad \widehat\mu_k = \frac{\mathbf{1}_{\mathrm{Box}_k}}{\mu_I(\mathrm{Box}_k)} \mu_I, \tag{36}$$

proved in (24) of [MB]. Its denominator is positive: $e_k < 1$ makes $p_{0,k} = (-\log e_k)^{p_0} > 0$, and the box has nonempty interior in the photon coordinate space.

All connections read by the step have their required domains. Indeed, scaling gives $\mathscr{A} \in \frac{1}{4}L^{-3/4}\widetilde{R}_k \subset \widetilde{R}_k^-$ and $\mathscr{C} \in \mathfrak{X}_k$. Theorem B of [MB] gives $A_k(\zeta) \in \mathrm{Dom}_k$ for every $\zeta \in \mathrm{Box}_k$. The real-gauge identity in (35) yields

$$H_k^{\mathrm{ph}} A_k(\zeta) = \mathscr{A} + \mathscr{Z}(\zeta) - d\omega_k.$$

Since $\frac{1}{4}\widetilde{R}_k$ is invariant under real gauges, $\mathscr{A} + \mathscr{Z}(\zeta)$ also lies in this domain. Thus both connection pairs $(\mathscr{A}, \mathscr{C})$ and $(\mathscr{A} + \mathscr{Z}, \mathscr{C})$ lie in $\widetilde{R}_k \times \mathfrak{X}_k$. By (19), $\mathscr{T}$, $\mathscr{T} + \mathscr{Z}$ and $\mathscr{T} + \mathscr{Z} - d\omega_k$, and their negatives, belong to $\widetilde{G}_k$. The fine and coarse inverse estimates consequently apply to them. The outgoing operators at $\mathscr{T}'$ then exist by (29).

For a total connection B let ev_B insert

$$\psi = H_k(B)\Psi, \qquad \bar{\psi} = \bar{H}_k(B)\bar{\Psi}, \qquad \chi = \delta_{\alpha,B} H_k(B)\Psi, \qquad \bar{\chi} = \delta_{\alpha,-B}\bar{H}_k(B)\bar{\Psi}.$$

At $k = 0$ the two field maps are identities. For complex $m, \varepsilon^{\mathrm{v}}$ and $\mathscr{E} \in K_k^{x,\mathbb{C}}$, with $\mathscr{E} = 0$ at $k = 0$, define

$$\begin{aligned} I_k^x[m, \mathscr{E}](A, C; \Psi) &= \mathrm{ev}_{A+C}\left(-m\langle\bar{\psi}, \psi\rangle + \sum_X \mathscr{E}(X; A, C)\right), \\ \mathrm{Act}_k^x[m, \mathscr{E}](A, C; \Psi) &= -\langle\bar{\Psi}, D_k(A + C)\Psi\rangle + I_k^x[m, \mathscr{E}](A, C; \Psi). \end{aligned} \tag{37}$$

The mass pairing is always the chamber pairing $\eta_k^3\, 48^{-1} \sum_{c,g,\beta} \bar{\psi}_\beta(c, g)\psi_\beta(c, g)$. The kinetic pairing uses the measure of $\mathscr{V}_k$. For an arbitrary finite family of scalar functions $\mathfrak{P}_k(X, C)$ on $\mathfrak{X}_k$, set

$$\mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}](A, C; \Psi) = \det D_k(0)^{-1} \exp\left\{\mathrm{Act}_k^x[m, \mathscr{E}](H_k^{\mathrm{ph}} A, C; \Psi) - \varepsilon^{\mathrm{v}} n_k + \sum_X \mathfrak{P}_k(X, C)\right\}. \tag{38}$$

The arguments here are $A \in \mathrm{Dom}_k$ and $C \in \mathfrak{X}_k$. The fields Ψ and $\bar{\Psi}$ are independent Grassmann variables.

Define the block transformation on an even density F whose coefficients are integrable along $A_k(\zeta)$ by

$$\begin{aligned} (\mathbf{T}_k^x F)(A', \mathscr{C}'; \Psi') &= c_k \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{D}\Psi\, e^{\mathcal{Q}_{\mathrm{bl}}} F(A_k(\zeta), \mathscr{C}; \Psi), \\ \mathcal{Q}_{\mathrm{bl}} &= -b_L\langle e^{-ie_k\omega_k}\bar{\Psi}'_L - \mathcal{A}_k(-\mathscr{B}_k - \mathscr{C})\bar{\Psi},\, e^{ie_k\omega_k}\Psi'_L - \mathcal{A}_k(\mathscr{B}_k + \mathscr{C})\Psi\rangle_L. \end{aligned} \tag{39}$$

The gauge function in the two outgoing phases is evaluated at the L-spaced sites. The Berezin functional is the weighted functional of (65) of [FB]; its incoming mode measure is 1 for $k \geq 1$ and $1/48$ for $k = 0$.

Its normalization is

$$\int \mathrm{D}\Psi'\, (\mathbf{T}_k^x F)(A', \mathscr{C}'; \Psi') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{D}\Psi\, F(A_k(\zeta), \mathscr{C}; \Psi). \tag{40}$$

To prove this, integrate the outgoing family first. Reciprocal multiplication of its barred and unbarred generators removes the two phases and leaves the top coefficient unchanged. Translation by the two averaged incoming fields likewise leaves it unchanged: every translated term that contains an external odd field has lower degree in the outgoing family. The remaining outgoing Gaussian exponent is $-b_L\langle\bar{\Psi}'_L, \Psi'_L\rangle_L = -b\langle\bar{\Psi}', \Psi'\rangle_1$. Its integral is $b^{4n_{k+1}} = c_k^{-1}$ by (68) of [FB]. The two Berezin integrations commute, since both extract an even top-degree coefficient; integration of the finite-dimensional coefficient functions in ζ also commutes with them. This proves (40).

For the density (38) the coefficient integrability follows from the domain statements above. The bond entries of all inverse operators are analytic on their invertibility domains, and the activity kernels are holomorphic. Composing them with the affine map $\zeta \mapsto H_k^{\mathrm{ph}} A_k(\zeta)$ gives continuous

coefficients on the compact photon box. Their even Grassmann exponentials are also continuous there.

Proposition 3.3. Let the data and domains be as above. Put

$$m' = m + m(\mathscr{E}), \qquad \mathscr{E}^r = R\mathscr{E}, \qquad m^+ = Lm', \qquad \varepsilon^{\mathrm{v},+} = L^3\big(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0\big).$$

At $k = 0$ the activity and its extracted terms vanish. Define the outgoing inserted fields $\psi' = H_{k+1}(\mathscr{T}')\Psi'$, $\bar\psi' = \bar H_{k+1}(\mathscr{T}')\bar\Psi'$ and their covariant Hölder quotients, and let $\psi^{0,\#}$ denote their scaled fine fields, including the barred ones. Let the means $\Psi^c, \bar\Psi^c$ be those of (31) at $B = \mathscr{T}$, and put

$$\Psi^f = \Psi^c + \Gamma_k(\mathscr{T})W, \qquad \bar\Psi^f = \bar\Psi^c + \bar W.$$

With $\mathscr{Z} = H_k^{\mathrm{ph}}\mathrm{Sq}_k\zeta$, set

$$\begin{aligned} V_k^x &= \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi^f) - \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^f) \\ &\quad + I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^f) - I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^c), \\ \Xi_k^x &= \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}_k(\zeta)\, e^{V_k^x}. \end{aligned} \tag{41}$$

Then the exact identity is

$$\begin{aligned} &(\mathbf{T}_k^x \mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}])(A', \mathscr{C}'; \Psi') \\ &\quad = \det D_{k+1}(0)^{-1} \exp\Big\{ - \langle \bar\Psi', D_{k+1}(\mathscr{T}')\Psi'\rangle_1 - m^+\langle \bar\psi', \psi'\rangle - \varepsilon^{\mathrm{v},+} n_{k+1} \\ &\qquad\qquad + \sum_X \mathscr{E}^r(X; \mathscr{A}, \mathscr{C}; \psi^{0,\#}) + \sum_X \mathfrak{P}_k(X, \mathscr{C})\Big\} \frac{\delta Z_k(\mathscr{T})}{\delta Z_k(0)} \Xi_k^x. \end{aligned} \tag{42}$$

All integrals in this formula are finite coefficient integrals. Its right side depends on A' through $H_{k+1}^{\mathrm{ph}}A'$. ∎

Proof. Extraction gives

$$I_k^x[m, \mathscr{E}](A, C; \Psi) = -\varepsilon_{\mathrm{v}}(\mathscr{E})n_k + I_k^x[m', \mathscr{E}^r](A, C; \Psi).$$

This is the image of (18) of [NC] under the field evaluation homomorphism. At $k = 0$ it holds with zero activity and $m' = m$.

For a fixed ζ, change the integrated variables by $\Psi \mapsto e^{ie_k\omega_k}\Psi$ and $\bar\Psi \mapsto e^{-ie_k\omega_k}\bar\Psi$. This leaves the weighted Berezin functional unchanged, since each mode's factors cancel in its top monomial, also for the $192n_0$ modes of weight $1/48$. The transport identity

$$U_{B - d\omega_k}(c, c')e^{ie_k\omega_k(c')} = e^{ie_k\omega_k(c)}U_B(c, c')$$

intertwines the Dirac operator and every average. Products and inverses preserve this intertwining, so it also holds for H_k and D_k. Its covariant difference identity uses the phase at the first photon site, including for a pair inside one cluster. The barred identity has the inverse phase. Gauge covariance of the activity and invariance of the mass pairing consequently turn its first connection argument $\mathscr{A} + \mathscr{Z} - d\omega_k$ into $\mathscr{A} + \mathscr{Z}$. The same change removes the gauge from the block Gaussian factor. The integral in (39) is now

$$\int \mathrm{D}\Psi\, \exp\Big\{-b_L\langle \bar\Psi'_L - \mathcal{A}_k(-\mathscr{T})\bar\Psi,\, \Psi'_L - \mathcal{A}_k(\mathscr{T})\Psi\rangle_L + \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi)\Big\},$$

multiplied by $\det D_k(0)^{-1}\exp\{-(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}))n_k + \sum_X \mathfrak{P}_k(X, \mathscr{C})\}$.

Separate the background quadratic term by writing

$$\begin{aligned} \mathcal{R}(\Psi) &= I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi) \\ &\quad + \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi) - \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi). \end{aligned}$$

The last exponent above is then the background Gaussian exponent plus $\mathcal{R}(\Psi)$. Translation by the means (31) and the block formula (71) of [FB] give

$$\delta Z_k(\mathscr{T}) e^{-\langle \bar\Psi', D_{k+1}(\mathscr{T}')\Psi'\rangle_1} \int \mathrm{d}\mu_I(W)\, e^{\mathcal{R}(\Psi^f)}.$$

The change of variables in this formula uses the same mode measure for Ψ and W. At $k=0$ it is the identity

$$\int \mathrm{D}\Psi\, e^{-\langle \bar\Psi, F_0(\mathscr{T})\Psi\rangle^{\mathrm{ch}}} G(\bar\Psi,\Psi) = \det F_0(\mathscr{T}) \int \mathrm{d}\mu_I(W)\, G(\bar W, \Gamma_0(\mathscr{T})W),$$

on the entire $192n_0$-mode space. The Gaussian measure is $e^{-\langle \bar W, W\rangle^{\mathrm{ch}}}\mathrm{D}W$, with the weight $\sqrt{48}$ in its coefficient bound.

By definition of V_k^x, $\mathcal{R}(\Psi^f) = I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^c) + V_k^x$. Its first summand is independent of W and ζ, so its even exponential factors out. The induced-field identity (32) gives

$$I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^c) = -m'\langle \bar\psi'_L, \psi'_L\rangle + \sum_X \mathscr{E}^r(X;\mathscr{A},\mathscr{C};\psi^{0,\#}).$$

The chamber measure changes by L^3 and each field by L^{-1}. Hence $\langle \bar\psi'_L, \psi'_L\rangle = L\langle \bar\psi', \psi'\rangle$, giving $m^+ = Lm'$. Normalizing the photon-box restriction in (36) adds $-\delta\varepsilon_k^0 n_k$ to the vacuum term. Since $n_k = L^3 n_{k+1}$, this term has exactly the stated outgoing vacuum coefficient. Finally (34) turns the remaining constant factor into $\det D_{k+1}(0)^{-1}\delta Z_k(0)^{-1}$, proving (42).

The fluctuation exponent is even and continuous on the photon box, by the same coefficient argument as for the input density. Thus its integral is defined, and its two finite integrations commute. Every background in the final expression is formed from $\mathscr{A}'$ and $\mathscr{C}'$; the remaining photon maps and measure are independent of A'. The minimum and the real gauge have disappeared from that expression, proving the last assertion. □

3.4. **Local substitution of the mean field.** We recall the result of [NC, Theorem B(iii)–(iv)] that is used between the two localizations. For $k \geq 1$ and unit generators over Y, the substitution fixes the old ordinary and pair generators and sets

$$\mathrm{sub}_{Y,B}\,\Psi = T_Y^{(1)}(B)\psi + T_Y^{(2)}(B)\chi, \qquad \mathrm{sub}_{Y,B}\,\bar\Psi = \bar T_Y^{(1)}(B)\bar\psi + \bar T_Y^{(2)}(B)\bar\chi. \tag{43}$$

The row measures are $\eta_k^3/48$ for ordinary columns and $(\eta_k^3/48)^2$ for pair columns. The rows add only ordinary legs over Y and boundary pairs $(x, x^{(3)})$ admissible for Y; they read only bonds meeting Y. The substitution preserves balance, parity and the lattice, local gauge, charge and reality identities of the old generators. In particular,

$$\begin{aligned} a_g\,\mathrm{sub}_{gY,gB} &= \mathrm{sub}_{Y,B}\,a_g, \\ G_\lambda\,\mathrm{sub}_{Y,B^1} &= \mathrm{sub}_{Y,B}\,G_\lambda, \qquad B^1 = B - d\lambda \text{ on bonds meeting } Y, \\ \mathrm{Cg}\,\mathrm{sub}_{Y,-B} &= \mathrm{sub}_{Y,B}\,\mathrm{Cg}, \\ \mathbf{K}\,\mathrm{sub}_{Y,-\overline{B}} &= \mathrm{sub}_{Y,B}\,\mathbf{K}. \end{aligned} \tag{44}$$

The local gauge equality need hold only on the bonds meeting Y. For $B \in \widetilde{R}_k + \mathfrak{X}_k$, the row bound gives

$$\|\,\mathrm{sub}_{Y,B}\,F\|_{w_1,w_2} \leq \|F\|_{w_1,w_2;w_3} \quad \text{if } C_T(w_1+w_2) \leq w_3. \tag{45}$$

The bounds on imaginary part and curvature in (20) verify the hypotheses of that result. The primitive inverse domain at B and $-B$ also verifies its additional recovery hypothesis. Therefore, on evaluation at $\psi = H_k(B)\Phi$, $\chi = \delta_{\alpha,B}\psi$ and the barred fields, the substitution sends $\Psi, \bar\Psi$ to $\Phi, \bar\Phi$. At zero no such substitution is needed.

4. The first chamber step

At $k=0$ write the charge as q. There is no input activity, and $H_0(B)=\bar H_0(B)=1$. Consequently the exponent in (41) is

$$V_0^x=-\langle\bar\Psi^f,(D_{\mathscr{T}+\mathscr{Z}}-D_{\mathscr{T}})\Psi^f\rangle^{\mathrm{ch}}-m\left(\langle\bar\Psi^f,\Psi^f\rangle^{\mathrm{ch}}-\langle\bar\Psi^c,\Psi^c\rangle^{\mathrm{ch}}\right). \tag{46}$$

The terms with μ and m cancel in the connection difference of the actions; the second difference of interactions gives the displayed mass term. Here $\Psi^c=\psi'_L$, $\bar\Psi^c=\bar\psi'_L$, $\Gamma_0(\mathscr{T})=F_0(\mathscr{T})^{-1}$, and $\Psi^f=\Psi^c+\Gamma_0(\mathscr{T})W$. All these are fields in $\mathscr{V}_0=\mathcal{S}^{\mathrm{ch}}(1,\ell)$. The identity measure in W has mode measure $1/48$, and its two-point kernel is 48 times the identity matrix of the modes.

For a bond fluctuation Z and an M-cube $\square$, define its local vertex by

$$\begin{aligned}\mathcal{V}_0(Z,\square)=\frac12\int_{\square}\mathrm{d}x\,\Big\{&\bar\psi(x)^TM_3(x)(e^{iqZ(c,c_3)}-1)[\psi(x)+\chi(x,x^{(3)})]\\&+[\bar\psi(x)+\bar\chi(x,x^{(3)})]^TM_3(x^{(3)})(e^{-iqZ(c,c_3)}-1)\psi(x)\Big\},\end{aligned} \tag{47}$$

where $x=(c,g)$, $c_3=c+ge_1$ and $\int_{\square}\mathrm{d}x=48^{-1}\sum_{c\in\square}\sum_g$. The pair $(x,x^{(3)})$ has $d_X=1$ and is admissible for $\square$ of the second kind. Thus the vertex has ordinary legs over the cube and precisely the boundary pair legs allowed in the activity algebra. Its coefficients are entire functions of Z on the bonds meeting the cube.

Proposition 4.1. Evaluate $\chi=\delta_{\alpha,\mathscr{T}}f$, $\bar\chi=\delta_{\alpha,-\mathscr{T}}\bar f$, $\psi=f$ and $\bar\psi=\bar f$ at any pair of antiperiodic odd chamber fields. Then

$$\mathrm{ev}\sum_{\square}\mathcal{V}_0(Z,\square)=-\langle\bar f,(D_{\mathscr{T}+Z}-D_{\mathscr{T}})f\rangle^{\mathrm{ch}}. \tag{48}$$

The two halves have equal images after summing over cubes. The vertex has the lattice, real-gauge, charge-conjugation and reality symmetries

$$\begin{aligned}a_g\mathcal{V}_0(gZ,g\square)&=\mathcal{V}_0(Z,\square),\\G_\lambda\mathcal{V}_0(Z,\square)&=\mathcal{V}_0(Z,\square),\\\mathrm{Cg}\mathcal{V}_0(-Z,\square)&=\mathcal{V}_0(Z,\square),\\\mathbf{K}\mathcal{V}_0(-\overline{Z},\square)&=\mathcal{V}_0(Z,\square).\end{aligned} \tag{49}$$

For any generator weights (w_1,w_2),

$$\|\mathcal{V}_0(Z,\square)\|_{w_1,w_2}\le 12(1+\rho)M^3w_1(w_1+w_2)\sup_{b\text{ oriented and meeting }\square}|e^{iqZ(b)}-1|. \tag{50}$$

If $q\le1$ and $q|Z(b)|\le\frac12q^{1-\varepsilon/2}$ on these bonds, then

$$\|\mathcal{V}_0(Z,\square)\|_{h_0/2}\le 6(1+\rho)M^3q^{1/2-\varepsilon/2}\le 12M^3q^{1/2-\varepsilon/2}. \tag{51}$$

■

Proof. At a nearest-neighbor pair the six paths reduce to the bond, so the quotient relations read

$$f(x)+\chi(x,x^{(3)})=u_{\mathscr{T}}(c,c_3)f(x^{(3)}),\qquad \bar f(x)+\bar\chi(x,x^{(3)})=u_{\mathscr{T}}(c,c_3)^{-1}\bar f(x^{(3)}).$$

The first half of the summed vertex is therefore one half of the integral of

$$\bar f(x)^TM_3(x)(e^{iqZ(c,c_3)}-1)u_{\mathscr{T}}(c,c_3)f(x^{(3)}).$$

Reindexing the second half by the measure-preserving involution $x\mapsto x^{(3)}$ gives the same integrand: both Z and $\mathscr{T}$ change sign on the reversed bond. Only the type-3 hop depends on the connection, and $u_{\mathscr{T}+Z}=e^{iqZ}u_{\mathscr{T}}$. The difference of the two Dirac operators is thus the negative of this hopping coefficient, proving (48). Every product contains two antiperiodic fields, so the reindexing is legitimate on the torus.

For lattice symmetry use $g^{-1}(x^{(3)}) = (g^{-1}x)^{(3)}$ and $S^{-1}M_3(x)S = M_3(g^{-1}x)$ for the spin lift S. Each half transforms separately, and reindexing maps $g\square \times O_{\mathrm{cub}}$ onto $\square \times O_{\mathrm{cub}}$ with the same measure. For a real gauge, the two generators in each term have the same first photon site and opposite bar indices, so their phases cancel. Charge conjugation interchanges the two halves. Its algebraic rule on the first half gives

$$\mathrm{Cg}[\bar\psi(x)^T m(\psi(x)+\chi(x,x^{(3)}))] = (\bar\psi(x)+\bar\chi(x,x^{(3)}))^T(\mathsf{C}^{-1}m^T\mathsf{C})\psi(x).$$

The sign from exchanging the odd factors cancels the minus sign in the image of $\bar\psi$. As

$$\mathsf{C}^{-1}M_3(x)^T\mathsf{C} = M_3(x^{(3)}),$$

the first half at $-Z$ becomes the second at Z; the reverse relation gives the other half. Finally

$$\varkappa^T\overline{M_3(x)}\varkappa = M_3(x), \qquad \overline{e^{\pm iq(-\overline{Z})}-1} = e^{\pm iqZ}-1.$$

The antilinear map consequently preserves each half in its reality identity. This proves (49).

Put $\delta_x = e^{iqZ(c,c_3)}-1$ and $\delta'_x = e^{-iqZ(c,c_3)}-1$. Collecting the ordinary bilinear in (47), its coefficient is $[M_3(x)\delta_x + M_3(x^{(3)})\delta'_x]/96$. The other two products have coefficients $M_3(x)\delta_x/96$ and $M_3(x^{(3)})\delta'_x/96$, and weights w_1w_2. The ordinary product has weight w_1^2. The chamber sites in the cube and the ordered pairs with these first sites are distinct, so these are exactly the coefficients, up to ordering signs. Each endpoint matrix satisfies $\sum_{\beta,\beta'}|M_3(\cdot)_{\beta\beta'}| \le 12(1+\rho)$, since $M_3 = 3(\rho \mp \gamma_\mu)$ and γ_μ has four entries of modulus one. There are $48M^3$ chamber sites in the cube. The triangle inequality for the collected bilinear therefore gives (50).

Under the smallness condition, $|e^{iqZ(b)}-1| \le 2q|Z(b)| \le q^{1-\varepsilon/2}$. For $w = h_0/2$, $w_1(w_1+w_2) = \frac14 q^{-1/2}(1+q^\varepsilon) \le \frac12 q^{-1/2}$. Inserting these bounds proves (51). □

For $Z = \mathscr{Z}$ and the evaluation at $\Psi^f, \bar\Psi^f$, the first term of (46) is the image of the summed vertex. Under the additional charge condition $96M^3q^{1/4} \le 1$, its bound becomes $\|\mathcal{V}_0(\mathscr{Z},\square)\|_{h_0/2} \le q^{1/4-5\varepsilon}$ whenever the displayed bond smallness condition holds. These statements apply before any expansion of the finite integral.

The first-step density identity retains every vacuum and determinant factor:

$$\begin{aligned}(\mathbf{T}_0^x\mathscr{F}_0^x[\varepsilon^{\mathrm{v}},m,0])(A',\mathscr{C}';\Psi') \\ = \det D_1(0)^{-1}\exp\Big\{ &-\langle\bar\Psi', D_1(\mathscr{T}')\Psi'\rangle_1 - Lm\langle\bar\psi',\psi'\rangle \\ &- L^3(\varepsilon^{\mathrm{v}}+\delta\varepsilon_0^0)n_1 + \sum_X \mathfrak{P}_0(X,\mathscr{C})\Big\} \\ \times \frac{\det F_0(\mathscr{T})}{\det F_0(0)}\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}_0(\zeta)\,e^{V_0^x}&.\end{aligned} \tag{52}$$

The two determinants of F_0 have size $192n_0$. The original level-zero density, integrated before the block transformation, is

$$\int \mathrm{D}\Psi\,\mathscr{F}_0^x[\varepsilon^{\mathrm{v}},m,0](A,C;\Psi) = e^{-\varepsilon^{\mathrm{v}}n_0+\sum_X\mathfrak{P}_0(X,C)}\frac{\det(D_{A+C}+\mu+m)}{\det D_0(0)}. \tag{53}$$

This follows directly from the weighted determinant identity. The numerator need not be invertible; the nonzero denominator uses $\mu > 0$.

5. Localizing the fluctuation activity

The finite fluctuation integral in (41) has a logarithm which is a sum of local activities. Its construction has two localizations. The first resolves the change of background and reconstructs the coarse mean; the second resolves the fermion fluctuations. After these finite operations, a

connected-polymer expansion gives the logarithm. The estimates below apply also to the first step, where the fluctuation generators have chamber measure 1/48.

5.1. **Contour domains and localization.** Keep the admissible sides, mass and charge of section 3, and assume in addition $\ell \geq Mr_k$. Put $e = e_k$, $h = (h_{k,1}, h_{k,2})$. The weakened photon and fermion primitives are taken on the polydisk

$$\mathrm{Pd}_k = \{s = (s', s'') : |s_{\mathrm{pc}}| \leq M^{\alpha_0}\}.$$

Here $s' = (s'_M, s'_{LM})$ contains the photon reconstruction and square-root parameters, of kinds π_M, π_{LM} in (37) of [MB]; s'' contains the fermion M-cube parameters π_k of Section 4.1 of [FB]. We assume their analytic kernel and Hölder estimates, component restriction, row vanishing and coefficient support with common constants $(\alpha_0, \gamma_w, \kappa_w; C_w)$, where $\kappa_w = \kappa + \kappa_0 + 3$. The photon hypotheses are those of Proposition 5.1 of [MB], (42) of [MB] and the component restriction following them. The fermion hypotheses are those of Propositions 4.1 of [FB] and 4.3 of [FB], including (57) of [FB], (58) of [FB], and their covariance identities. At parameter value one all families equal the exact operators. Their gauge, lattice, transpose and reality actions are the actions of those exact operators with the parameters transformed accordingly. At level zero we retain the stated diagonal families and the chamber covariance with measure 1/48.

The quantitative assumptions are those of theorem A.

For $\zeta \in \mathrm{Box}_k$, write

$$Z(s') = H_k^{\mathrm{ph}}(s') C C_k^{1/2}(s')\zeta. \tag{54}$$

In this formula C is the real photon coordinate map of (21) of [MB]. The diagonal photon family is used at $k = 0$. For a bond field Z set

$$|Z|_* = \max\{\|Z\|_\infty, \|\partial Z\|_\infty, \|\delta_\alpha \partial Z\|_\infty\}.$$

The three thresholds defining R_k are $\theta_i = e^{-3/4+i\varepsilon}$, $i = 1, 2, 3$. Thus $\widetilde{R}_k^-$ consists of the permitted real-gauge part plus a field satisfying the three strict bounds $(1 - e^{\varepsilon})\theta_i$. Introduce the radii

$$r_1 = \tfrac{1}{2} e^{-3/4+9\varepsilon/2}, \qquad r_2 = e^{-3/4+7\varepsilon}, \qquad r_4 = e^{-1/4+3\varepsilon}. \tag{55}$$

Proposition 5.1. For every $s \in \mathrm{Pd}_k$ and $\zeta \in \mathrm{Box}_k$,

$$|Z(s')|_* \leq \tfrac{1}{2} e^{-\varepsilon/2}.$$

The radii satisfy $2 \leq r_2, r_4 \leq r_1$ and $(r_i - 1)^{-1} \leq 2/r_i$, $i = 2, 4$. If $A \in \widetilde{R}_k^-$, $\mathscr{C} \in \mathfrak{X}_k$, and $|Z|_* \leq \frac{1}{2} e^{-\varepsilon/2}$, then

$$A + tZ \in \widetilde{R}_k, \qquad A + \mathscr{C} + tZ \in \widetilde{\mathcal{G}}_k \quad (|t| \leq r_1).$$

■

Proof. The square-root kernel is bounded by $C_w \exp(-\gamma_w d(\Upsilon, \Upsilon'))$. The summability constant c_γ and the coordinate bound $|\zeta(\Upsilon')| \leq p_{0,k} \leq e^{-\varepsilon/4}$ give

$$\sup_{\Upsilon} |(C_k^{1/2}(s')\zeta)(\Upsilon)| \leq C_w c_\gamma e^{-\varepsilon/4}.$$

The coordinate map is real. Its bound $\|Cv\|_\infty \leq c_C \sup_\Upsilon |v(\Upsilon)|$ also gives the absolute row bound $\sum_\Upsilon |C(b, \Upsilon)| \leq c_C$: apply the real bound to the signs of the entries of a fixed row. Therefore

$$\|C C_k^{1/2}(s')\zeta\|_\infty \leq c_C C_w c_\gamma e^{-\varepsilon/4}.$$

Each of the reconstruction, derivative and Hölder derivative kernels has the same summable exponential bound. Consequently

$$|Z(s')|_* \leq c_C C_w^2 c_\gamma^2 e^{-\varepsilon/4} \leq \tfrac{1}{2} e^{-\varepsilon/2},$$

where the last inequality is $2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \le 1$. At level zero the polynomial diagonal family has the same row bounds: its constant is $4\exp(2\gamma_w)M^{27\alpha_0} \le C_w$.

Since $\varepsilon \le 1/200$,

$$\frac{r_2}{r_1} = 2e^{5\varepsilon/2} \le 1, \qquad \frac{r_4}{r_1} = 2e^{1/2-3\varepsilon/2} \le 2e^{5\varepsilon/2} \le 1.$$

The small-charge condition $88M^3e^{1/4-6\varepsilon} \le 1$ gives $r_4^{-1} = e^{1/4-3\varepsilon} \le 1/88$ and $r_2^{-1} \le r_4^{-1}$. In particular both radii are at least 2, and their stated inverse bounds follow.

For $|t| \le r_1$ the three norms of tZ are at most $\frac{1}{4}e^{-3/4+4\varepsilon}$, strictly less than $e^{\varepsilon}\theta_3$. Adding this field fills less than the margin in each of the three defining bounds of $\widetilde{R}_k^-$. The permitted gauge part is unchanged. Hence $A + tZ \in \widetilde{R}_k$. Finally $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{\mathcal{G}}_k$ gives the second inclusion. □

The first-step vertex estimate (51) now applies to $Z(s')$ for every weakening parameter. In particular, $96M^3e^{1/4} \le 1$ gives its bound $e^{1/4-5\varepsilon}$ in the $h/2$ norm.

Let $B = A + \mathscr{C}$. The arguments of $H_k^\#$ are independent pairs $(\Phi, \bar\Phi)$; in $H_k^\#\Phi$ the single symbol denotes this pair. Write $\psi^\# = (\psi, \chi; \bar\psi, \bar\chi)$ for the ordinary, pair, barred ordinary and barred pair generators, respectively. Then

$$\begin{aligned} H_k^\#(s'', B)\Phi = (&H_k(s'', B)\Phi, \delta_{\alpha,B}H_k(s'', B)\Phi; \\ &\bar H_k(s'', B)\bar\Phi, \delta_{\alpha,-B}\bar H_k(s'', B)\bar\Phi), \\ \bar H_k(s'', B) &= \mathsf{C}^{-1}H_k(s'', -B)\mathsf{C}. \end{aligned}$$

Define $J_k^\#(s'', B, Z) = H_k^\#(s'', B + Z) - H_k^\#(s'', B)$ and

$$\widetilde\Gamma_k(s'', B)(W, \bar W) = (\Gamma_k(s'', B)W, \bar W), \qquad W_k^\#(s'', B) = H_k^\#(s'', B)(\Gamma_k(s'', B)W, \bar W).$$

The notation $J_k, \bar J_k$ denotes the components without Hölder quotient; $\mathrm{Wc}, \overline{\mathrm{Wc}}$ denotes the corresponding components of $W_k^\#$. The auxiliary generators W have weight $\mathsf{w}_W = 1$ for $k \ge 1$ and $\sqrt{48}$ for $k = 0$.

These fields are continuous in all closed polydisk and box variables and analytic in the open parameter polydisk and the connection variables. For the Hölder rows this follows directly from

$$(\delta_{\alpha,B}H_k)((x, x'), u) = d_X(x, x')^{-\alpha}\big(U_B(x, x')H_k(x', u) - H_k(x, u)\big).$$

The transport is entire in B. Proposition 5.1 places every connection on the closed contours inside the open domain of the primitives. Thus compositions at $B + tZ(s')$ have the stated regularity also when the contour variable is on its circle.

In the following localization, cubes and polymers are sets of photon sites. Containment refers to all their lattice sites, and two cubes meet when they share a lattice site. Let $P_0(X)$ be the parameter cubes contained in X. For a set P of parameter cubes disjoint from $P_0(X)$, let $Y'(X, P)$ be the union of the M-cubes meeting X or a cube of $P_0(X) \cup P$. For a continuous analytic family $G(s, X)$ define

$$\mathcal{T}_P G(X) = \int_{[0,1]^P} \mathrm{d}s_P\, \partial_P G(s, X)\big|_{s=1 \text{ on } P_0(X),\, s=0 \text{ outside } P_0(X)\cup P}, \tag{56}$$

$$\mathrm{Loc}\, G(Y) = \sum_{(X,P):Y'(X,P)=Y} \mathcal{T}_P G(X). \tag{57}$$

Repeated one-variable integration gives the useful identity

$$\mathcal{T}_P G(X) = \sum_{Q\subseteq P} (-1)^{|P|-|Q|} G(\mathbf{1}_{P_0(X)\cup Q}, X). \tag{58}$$

In particular, coefficient locality can be checked at indicator families; no statement about the convergence of the later logarithm is used here.

Lemma 5.2. Suppose $\|G(s,X)\| \le B_0 \exp(-\kappa' d_M(X))$ on the parameter polydisk, where $\kappa_0 \le \kappa' \le \kappa$. Assume $\mathcal{T}_P G(X) = 0$ whenever $Y'(X,P)$ is disconnected. Then

$$\sum_Y \operatorname{Loc} G(Y) = \sum_X G(1,X), \tag{59}$$

$$\|\operatorname{Loc} G(Y)\| \le 4 c_X K_0 B_0 \exp\big(-(\kappa' - \kappa_0 - 1) d_M(Y)\big). \tag{60}$$

■

Proof. Applying $f(1) = f(0) + \int_0^1 f'(t)\mathrm{d}t$ in each parameter outside $P_0(X)$ gives $G(1,X) = \sum_P \mathcal{T}_P G(X)$. The sums are finite. After removing the vanishing disconnected terms, each term belongs to exactly one polymer $Y'(X,P)$. This proves (59).

Put $R = M^{\alpha_0}$. Cauchy's estimate on a polydisk of radius less than $R-1$ about each integration point, followed by the limit of these radii to $R-1$, gives

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-\kappa' d_M(X))(R-1)^{-|P|}.$$

If $Y'(X,P) = Y$, then $X \subseteq Y$ and $|Y \setminus X|_M \le N_c|P|$, since every added M-cube meets a parameter cube and each parameter cube meets at most $N_c = (2L+1)^3$ such cubes. To compare their tree lengths, order the cubes of $Y \setminus X$ by graph distance from X. Each added cube is adjacent to a preceding one. Extend a net by a segment from a point of that preceding closed cube to their common face, of length at most M. Starting from an arbitrarily near-minimal net for X gives $d_M(Y) \le d_M(X) + |Y \setminus X|_M$. Conversely every net for Y is a net for X. Thus $0 \le d_M(Y) - d_M(X) \le N_c|P|$. Writing $a_X = \log(6c_X)$, the condition $R - 1 \ge \exp(N_c\kappa + a_X)$ implies

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-(\kappa' - \kappa_0) d_M(Y)) \exp(-\kappa_0 d_M(X)) \exp(-a_X|P|).$$

At most $3|Y|_M$ parameter cubes are contained in Y. Hence

$$\sum_P \exp(-a_X|P|) \le (1 + (6c_X)^{-1})^{3|Y|_M} \le \exp((d_M(Y)+1)/2).$$

The polymer counting and size estimates give

$$\sum_{X \subseteq Y} \exp(-\kappa_0 d_M(X)) \le K_0^{|Y|_M} \le 2 c_X K_0 \exp(d_M(Y)/2).$$

The product of the last two bounds is at most $4 c_X K_0 \exp(d_M(Y))$. Summing the estimate for $\mathcal{T}_P G(X)$ proves (60). □

5.2. **The two finite localizations.** For a function analytic near the closed disk $|t| \le r$, $r > 1$, write

$$\mathcal{C}_r f = \frac{1}{2\pi i} \int_{|t|=r} \frac{f(t)}{t(t-1)}\,\mathrm{d}t.$$

This is $f(1) - f(0)$ and satisfies $\|\mathcal{C}_r f\| \le (r-1)^{-1} \sup_{|t|=r} \|f(t)\|$. Let $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(4)$ and set $m' = m + m(\mathscr{E})$, $\mathscr{E}^r = R\mathscr{E}$. For $k \ge 1$ define

$$G_1(s,X) = \mathcal{C}_{r_1} \langle \bar\Psi, M_k(X, B + tZ(s'))\Psi\rangle_1, \tag{61}$$

$$G_2(s,X) = \mathcal{C}_{r_2}\, \mathscr{E}^r(X; A + tZ(s'), \mathscr{C}, \psi^\# + tJ_k^\#(s'', B, Z(s'))\Psi), \tag{62}$$

$$G_3(s,\square) = \mathcal{C}_{r_2}(-m')\langle \bar\psi + t\bar J_k(s'', B, Z(s'))\bar\Psi, \psi + tJ_k(s'', B, Z(s'))\Psi\rangle_\square. \tag{63}$$

The symbols $M_k(X,\cdot)$ denote the local Dirac primitives. A function defined for one cube is zero on the other polymers. Put

$$E_i^*(Y) = \operatorname{Loc} G_i(Y), \qquad E_i(Y) = \operatorname{sub}_{Y,B} E_i^*(Y), \qquad E_{\le 3} = \sum_{i=1}^3 E_i.$$

The substitution is (43). At $k = 0$ there is no Ψ and no substitution: $G_1(s, \square) = \mathcal{V}_0(Z(s'), \square)$, $G_2 = G_3 = 0$, and $E_{\leq 3} = \operatorname{Loc} G_1$. All parameter kinds occur in this first localization.

For the second localization, which uses only π_k, let $v = W_k^{\#}(s'', B)$ and define

$$G^W_{\leq 3}(s'', X) = E_{\leq 3}(X; A, \mathscr{C}, \zeta, \psi^{\#} + v), \tag{64}$$

$$G_4(s'', X) = \mathcal{C}_{r_4} \mathscr{E}^r(X; A, \mathscr{C}, \psi^{\#} + tv), \tag{65}$$

$$G_5(s'', \square) = \mathcal{C}_{r_4}(-m')\langle \bar{\psi} + t\overline{\mathrm{Wc}}, \psi + t\mathrm{Wc}\rangle_{\square}, \tag{66}$$

$$F(Y) = \operatorname{Loc}(G^W_{\leq 3} + G_4 + G_5)(Y). \tag{67}$$

At $k = 0$ the term G_4 is zero and $m' = m$.

Proposition 5.3. All finite expressions just defined exist on $\widetilde{R}_k^- \times \mathfrak{X}_k \times \mathrm{Box}_k$. The element $F(Y)$ is even, has ordinary legs over Y and pair legs admissible for Y, and has fluctuation generators only at unit indices in Y. Its coefficients read $A, \mathscr{C}$ only on bonds meeting Y and read ζ only through the coordinates $\zeta(\Upsilon)$ with $\operatorname{supp} \Upsilon \subset Y$. They are continuous in these arguments and analytic in the connection variables. The same assertions hold for $E_{\leq 3}(Y)$, which has no fluctuation generators.

The disconnected terms in both localizations vanish, and

$$\sum_Y E_i^*(Y) = \sum_X G_i(1, X), \qquad \sum_Y F(Y) = \sum_X (G^W_{\leq 3} + G_4 + G_5)(1, X). \tag{68}$$

The finite expressions on Y depend on the datum only through m' and the $\mathscr{E}^r(X)$ with $X \subseteq Y$. ■

Proof. We give the component argument which is responsible for locality. Let $\mathrm{Ps} = P_0(X) \cup P$, let $Y' = Y'(X, P)$ and let Y'_1 be its component containing X. Set the weakening parameters to zero outside Ps. For each of the three parameter kinds, any link-component Lnk of Ps is contained in the polymer formed by the M-cubes meeting it. This polymer lies in one component of Y'. If a site of Y'_1 lies in Lnk, or in the enlarged parameter cube of a photon kind, that polymer has a cube in common with Y'_1, and hence lies entirely in Y'_1.

The row-vanishing property of the weakened photon families and their component restriction give, on any fine bond b meeting Y'_1,

$$Z(s')(b) = \sum_{b', \Upsilon, \Upsilon' \subset Y'_1} H_k^{\mathrm{ph}}(s')(b, b') C(b', \Upsilon) C_k^{1/2}(s')(\Upsilon, \Upsilon') \zeta(\Upsilon'). \tag{69}$$

Indeed, the reconstruction row is zero unless b lies in one link-component; its columns lie in that component. The coordinate map has columns supported in the L-blocks of the ends of b', and each block lies in the M-cube of the corresponding end. The square-root row then has columns in its own link-component. Each component involved is attached to Y'_1 by the preceding argument. These rows are unchanged if the parameters outside Y'_1 are changed while their common zero set outside Ps is kept.

For a chamber site $x = (c, g)$ at distance at most 1 from Y'_1, put $K_c = \{c' : |c' - c|_\infty \leq 2L\}$, as in [FB, Proposition 4.1]. The fermion row is zero unless every parameter cube meeting K_c is present. These cubes form one link-component. When the row is nonzero, that component contains c and a site of Y'_1, so it lies in Y'_1. The row of $H_k(s'', B)$, and its barred counterpart, then has columns in that component and reads B only on bonds meeting it. At a column of this row, the corresponding row of $\Gamma_k(s'', B)$ has columns in the same component. These conclusions also hold at level zero: the diagonal factor is

$$\prod_{\mathrm{pc} \in \pi_0 : \mathrm{pc} \cap K_c \neq \varnothing} s''_{\mathrm{pc}},$$

and the kernel of that multiplication operator is $48\mathbf{1}_{x=x'}$ times this factor. Its composition with Γ_0 uses the measure $1/48$.

An endpoint of a pair admissible for X is at distance at most $\eta_k \le 1$ from X. Its transport reads only bonds contained in X, or the one type-3 bond meeting X. Thus the preceding row argument applies to every row used in $J_k^{\#}$ and $W_k^{\#}$. The Hölder row has only the two columns x, x' and the entire transport coefficient displayed above. Finite compositions preserve this dependence: the columns of O_2O_1 are the union of the columns of O_1 at the finitely many columns of O_2, and its read bonds are the union of their read bonds. Entries of the transport and reconstruction rows are entire in the connections, and finite compositions of these algebraic rows retain this property. Compositions involving weakened primitives are holomorphic on their stated primitive domains. These are the finite-row properties used here; the weakened operators themselves are not asserted to have finite range.

For G_1 at $k \ge 1$, the local primitive $M_k(X, \cdot)$ has both indices in X and reads only bonds meeting X. For G_2, G_3, the incoming activity or mass form has ordinary legs in X and admissible pair legs; the substitution by $J_k^{\#}\Psi$ introduces Ψ only in Y_1'. At $k = 0$ the vertex has precisely these ordinary and type-3 pair legs and reads Z only on bonds meeting its cube. Equation (69) therefore proves the same component dependence for every first-localization function.

If Y' is disconnected, choose an M-cube outside Y_1'. It cannot meet X or a parameter cube contained in X, so it meets a cube of P outside Y_1'. The integrand is independent of that parameter. Pairing the subsets in (58) according to whether they contain this cube proves that the term is zero. For a term assigned to a connected Y, all parameter cubes in $P_0(X) \cup P$ and X lie in Y. Equation (58) now gives the claimed locality of E_i^*.

The reconstruction substitution adds only ordinary legs over Y and boundary pairs $(x, x^{(3)})$ admissible for Y. Its finite rows at a unit site y read bonds contained in Δ_y or type-3 bonds from this block. These bonds meet Y since $\Delta_y \subset Y$. The barred rows have the same support. Thus it preserves the claimed coefficient locality and removes Ψ. For the second localization, the row argument for $H_k^{\#}\widetilde{\Gamma}_k W$ gives fluctuation generators only in Y_1'. It applies to each of the three functions in (67). The same disconnected parameter cancellation proves the second assertion. Lemma 5.2 gives both sum identities.

Regularity follows from the established contour domains, the primitive regularity, and finite sums and substitutions. Contour integrals preserve analyticity: their Riemann sums converge uniformly on compact sets. The same argument applies to finite parameter integrals. The incoming activities and mass forms are even, and each substitution sends generators to odd linear combinations, so every expression remains even. Finally G_1 is independent of the datum; G_2, G_4 use $\mathscr{E}^{er}(X)$ and G_3, G_5 use m'. Tracing the two localizations shows that no incoming polymer outside Y enters the result. □

The proof identifies the primitive entries determining a finite expression on Y. With parameters zero outside Y, the photon field reads only $H_k^{\mathrm{ph}}(s')(b, b')$ for b meeting Y, $b' \subset Y$, and $C_k^{1/2}(s')(\Upsilon, \Upsilon')$ with both supports in Y. Its remaining factor is the fixed coordinate map. The fermion fields use $H_k, \bar{H}_k$ at chamber sites over Y, with unit columns in Y, at $B + Z(s')$ and B, and $\Gamma_k(s'', B)$ with both unit indices in Y. At zero, the kernel sums have measure $1/48$ and the diagonal rows have the factor 48 already recorded. The Dirac primitive has both indices in $X \subseteq Y$ and connection $B + tZ(s')$ on $|t| = r_1$. Hölder transports read only bonds meeting Y. The reconstruction substitution contains only the fixed finite $T_k^{(i)}, \bar{T}_k^{(i)}$ rows at unit sites of its own subpolymer.

Consequently two systems with the stated hypotheses on this torus give the same finite expressions on Y if these entries agree for every parameter family zero outside Y, at the connections formed with their respective photon fields. The restricted photon sum first gives equality of the fields on bonds meeting Y; the transports and incoming activities there then agree by their locality. The listed fermion and Dirac entries give equal first functions, their localizations and reconstruction substitutions. They give equal second functions and localizations in turn, and then equal activities $K(Y)$ and $H(Y)$ in (78) and (79), and each term of the connected series (80). Agreement of the

photon entries and agreement of the fermion entries as functions on the full operator domain is sufficient: if the two composed connections differ outside Y, local row dependence removes that difference first. The same comparison passes to the convergent logarithm once its convergence has been established.

5.3. **Bounds before fluctuation integration.** Set

$$\kappa_w = \kappa + \kappa_0 + 3, \qquad a_1 = \kappa - \kappa_0 - 1, \qquad a_2 = \kappa - 2\kappa_0 - 2,$$
$$C_W = 28C_w^2 c_\gamma, \qquad H_{\mathrm{cl}} = 36 c_X^2 K_0^2 e^{1/4-5\varepsilon}.$$

The notation H_{cl} is a scalar cluster bound. For a scalar-component kernel $O(a,b)$ with column measure $\nu(b)$, write $|O|_{1,\infty} = \sup_a \sum_b \nu(b)|O(a,b)|$, including spin indices. This is the row norm of [NC, Theorem B]. The finite primitive bounds imply

$$\|\langle \bar\Psi, M_k(X,B)\Psi\rangle_1\|_{w_3} \le 4c_X c_\gamma C_w M^3 w_3^2 \exp(-(\kappa_w - 1)d_M(X)), \quad k \ge 1, \tag{70}$$
$$|H_k^{\#}(s'',B)|_{1,\infty} \le 4C_w, \tag{71}$$
$$|J_k^{\#}(s'',B,Z)|_{1,\infty} \le 16 C_w e^{3/4-9\varepsilon/2}, \tag{72}$$
$$\mathsf{w}_W |H_k^{\#}(s'',B)\widetilde\Gamma_k(s'',B)|_{1,\infty} \le C_W. \tag{73}$$

Here $B = A + \mathscr{C}$ and $|Z|_* \le \frac12 e^{-\varepsilon/2}$. To see the first bound, count $4M^3|X|_M$ rows and sum the exponential kernel in its second index, giving $c_\gamma C_w \exp(-\kappa_w d_M(X))$ per row. Then $|X|_M \le c_X \exp(d_M(X))$ gives (70). For $H_k^{\#}$ the primitive bound controls the ordinary and short-pair rows by C_w. A long pair, $d_X \ge 1$, has its Hölder quotient bounded by four times the ordinary sup norm. At level zero every pair is long. The monomial matrices $\mathsf{C}^{\pm1}$ preserve row norms, so the barred rows have the same bounds. Apply $\mathcal{C}_{r_1}$ to $H_k^{\#}(s'', B + tZ)$ to obtain the bound for $J_k^{\#}$. Finally $|\Gamma_k|_{1,\infty} \le C_w c_\gamma$ by kernel summation, using $1/48$ at level zero. Submultiplicativity and $\mathsf{w}_W \le 7$ give (73).

Proposition 5.4. For the data of proposition 5.3,

$$\|E_{\le 3}(Y)\|_{h/2} \le 8 c_X K_0 e^{1/4-5\varepsilon} \exp(-a_1 d_M(Y)), \tag{74}$$
$$\|F(Y)\|_{(h/4;\mathsf{w}_W)} \le H_{\mathrm{cl}} \exp(-a_2 d_M(Y)). \tag{75}$$

The element $F(Y)$ is the sum of a term independent of $(m, \mathscr{E})$ and a term complex linear in these data. ∎

Proof. Extraction gives

$$\|\mathscr{E}^r(X)\|_h \le 2\|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)) \le 12 e^{1/4-7\varepsilon} \exp(-\kappa d_M(X)), \qquad |m'| \le 8e^{3/4-8\varepsilon}.$$

The mass bound uses $c_{\mathfrak{fl}} e^{\varepsilon} \le 1$. With chamber measure $\eta_k^3/48$, direct counting gives

$$\|\langle \bar\psi, \psi\rangle_\square\|_{(w_1,w_2)} = 4M^3 w_1^2.$$

The number of monomials is $4 \cdot 48 \eta_k^{-3} M^3$ and their common coefficient is $\eta_k^3/48$.

In the first localization give Ψ the weight $w_3 = C_T h_{k,1}$ and the old generators the weights $h/2$. The contour bound and (70) give

$$\|G_1(s,X)\| \le 16 c_X c_\gamma C_w C_T^2 M^3 e^{1/4-9\varepsilon/2} \exp(-(\kappa_w - 1)d_M(X)) \le e^{1/4-5\varepsilon} \exp(-\kappa d_M(X)).$$

The last step uses

$$16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \le 1.$$

At $k = 0$ the weighted vertex instead gives $12M^3 e^{1/2-\varepsilon/2} \le e^{1/4-5\varepsilon}$. Moreover,

$$r_2 |J_k^{\#}|_{1,\infty} w_3 \le 16 C_w C_T e^{3\varepsilon/2} h_{k,2} \le \tfrac12 h_{k,2}.$$

The substitution bound therefore takes the integrand norm of G_2 to the h norm of $\mathscr{E}^r$, and that of G_3 to the h norm of the mass form. It gives

$$\|G_2(s,X)\| \le 24e^{1-14\varepsilon}\exp(-\kappa d_M(X)), \qquad \|G_3(s,\square)\| \le 64M^3e^{1-15\varepsilon}.$$

The condition $88M^3e^{1/4-6\varepsilon} \le 1$ implies

$$24e^{3/4-9\varepsilon} + 64M^3e^{3/4-10\varepsilon} \le 1.$$

Thus $\sum_{i\le 3} G_i$ is bounded by $2e^{1/4-5\varepsilon}\exp(-\kappa d_M(X))$. Apply lemma 5.2, using the disconnected cancellation in proposition 5.3. The reconstruction substitution costs no norm, since $C_T(h_{k,1}+h_{k,2})/2 \le w_3$. This proves (74).

For the second localization, the condition $112C_w^2c_\gamma e^{2\varepsilon} \le 3$ gives

$$r_4C_W \le \tfrac{3}{4}h_{k,2}, \qquad C_W \le \tfrac{1}{4}h_{k,2}.$$

Consequently addition of $W_k^{\#}$ maps the $h/4$ norm to the $h/2$ norm at $t=1$ and to the h norm on the contour $|t|=r_4$. The three input bounds are

$$\begin{aligned}\|G^W_{\le 3}(s'',X)\| &\le 8c_XK_0e^{1/4-5\varepsilon}\exp(-a_1d_M(X)),\\ \|G_4(s'',X)\| &\le 24e^{1/2-10\varepsilon}\exp(-\kappa d_M(X)),\\ \|G_5(s'',\square)\| &\le 64M^3e^{1/2-11\varepsilon}.\end{aligned}$$

Their sum is at most $9c_XK_0e^{1/4-5\varepsilon}\exp(-a_1d_M(X))$, again by $88M^3e^{1/4-6\varepsilon}\le 1$. Since $\kappa_0 \le a_1 \le \kappa$, resummation gives (75). At level zero $G_4 = 0$; all the other inequalities include the factor w_W already, so they apply unchanged. Linearity follows from the definitions: G_1 is fixed, G_2, G_4 are linear in $R\mathscr{E}$, G_3, G_5 in m', and localization, the reconstruction substitution and the contour functionals are linear. □

6. The connected logarithm

Two polymers are incompatible when their union is connected through faces. In particular a polymer is incompatible with itself. For an ordered tuple $(X_1,\dots,X_n)$ define $u^T(X_1)=1$ and

$$u^T(X_1,\dots,X_n) = \sum_{G \text{ connected}} (-1)^{|G|}, \quad n \ge 2,$$

where G ranges over spanning subgraphs of the incompatibility graph of the tuple. The coefficient is zero when that graph is disconnected.

We will use the formal connected-graph identity

$$\sum_{\{X_i\}\text{ compatible}} \prod_i z_{X_i} = \exp\left\{\sum_{n\ge 1}\frac{1}{n!}\sum_{(X_1,\dots,X_n)} u^T(X_1,\dots,X_n)\prod_i z_{X_i}\right\} \tag{76}$$

and the tree bound

$$|u^T(X_1,\dots,X_n)| \le \#\{\text{spanning trees of the incompatibility graph}\}. \tag{77}$$

Here the identity is one of formal series in commuting variables, before substitution into a normed algebra. For its proof, express the left side as an ordered sum with factor $1/n!$ and multiply by $\prod_{i<j}(1-\mathbf{1}_{\{X_i,X_j\}\text{ incompatible}})$. Expanding this product sums over all graphs. A graph is uniquely the union of its connected components. Grouping by their vertex sets gives the exponential on the right: an ordered list of component sizes $n_1,\dots,n_r$ has $n!/(n_1!\cdots n_r!)$ vertex assignments, and division by $r!$ removes the order of its components. Only finitely many terms contribute to a fixed degree.

For (77), let $\mathbf{G}$ be the incompatibility graph and suppose it is connected. For a connected spanning subgraph G, root at 1 and, for $i \ne 1$, choose as its parent the least labelled neighbor at

distance $d_G(i)-1$ from the root. These edges form a tree $T(G)$ with the same root distances as G. Given a spanning tree τ, let $R(\tau)$ contain its edges, all edges of $\mathbf{G}$ within one distance level, and all edges from i to the preceding level whose other endpoint exceeds the parent of i in τ. Then

$$T(G)=\tau \quad\Longleftrightarrow\quad \tau\subseteq G\subseteq R(\tau).$$

In one direction the distances and least parents force every edge of G to belong to $R(\tau)$. Conversely, along any edge of $R(\tau)$ the tree distance changes by at most one. Thus a graph between τ and $R(\tau)$ has the tree's distances, and the least neighbor in the preceding level is its tree parent. The connected subgraphs are therefore partitioned into these intervals, and their signed sum is

$$\sum_{\tau}(-1)^{n-1}\sum_{A\subseteq R(\tau)\setminus\tau}(-1)^{|A|}=(-1)^{n-1}\#\{\tau : R(\tau)=\tau\}.$$

This proves the tree bound, including the cases of a disconnected graph and $n=1$.

For this section and the symmetry proof, write $\widehat{\mu}=\widehat{\mu}_k$ for the normalized product box law in (36).

Proposition 6.1. Let $V(X)$ be even elements in the old and fluctuation generators, continuous on Box_k, with

$$\|V(X)\|_{(w;\mathsf{w}_W)}\le H_{\mathrm{cl}}\exp(-a_2 d_M(X)).$$

Assume that $V(X)$ has fluctuation indices only in X and depends on ζ only through coordinates supported in X. For any union Y of M-cubes put

$$K(Y)=\sum_{\substack{\{X_i\}\text{ distinct}\\ \bigcup_i X_i=Y}}\prod_i(\exp V(X_i)-1),\qquad K(\varnothing)=1, \tag{78}$$

$$H(Y)=\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)\,K(Y). \tag{79}$$

For a connected Y define

$$E^{\#}(Y)=\sum_{n\ge1}\frac{1}{n!}\sum_{\bigcup_i Y_i=Y}u^T(Y_1,\dots,Y_n)\prod_i H(Y_i). \tag{80}$$

The last series converges absolutely in the w norm and

$$\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)\exp\left\{\sum_X V(X)\right\}=\exp\left\{\sum_Y E^{\#}(Y)\right\}, \tag{81}$$

$$\|E^{\#}(Y)\|_w\le c'_S H_{\mathrm{cl}}\exp(-a_3 d_M(Y)), \tag{82}$$

Here $a_3=\kappa-5\kappa_0-4$ and $c'_S=14c_X^2K_0^2\exp(3\kappa)$. The sum of the norms of the terms with $n>N$ in (80) is at most 2^{-N} times the right side of (82). ∎

Proof. Set

$$a_4=\kappa-3\kappa_0-3,\quad a_5=\kappa-3\kappa_0-2,\quad a_6=\kappa-5\kappa_0-3,$$
$$H_1=7c_XK_0\exp(3a_5)H_{\mathrm{cl}},\qquad z_S=28H_1\exp(\kappa_0)K_0\exp(3a_6).$$

The fixed small-charge choice

$$c_Se^{1/4-5\varepsilon}\le1,\qquad c_S=10^5c_X^5K_0^5\exp(6\kappa)$$

implies

$$H_{\mathrm{cl}}\le1,\qquad 4H_{\mathrm{cl}}K_0c_X\exp(3a_5)\le1,\qquad z_S\le\tfrac12. \tag{83}$$

Indeed the second left side is $144c_X^3K_0^3\exp(3a_5)e^{1/4-5\varepsilon}$. For the third, $2z_S=14112c_X^3K_0^4\exp(3a_5+3a_6+\kappa_0)e^{1/4-5\varepsilon}$, and $3a_5+3a_6+\kappa_0\le6\kappa$.

Even elements commute, so expanding their exponential gives $\exp(\sum_X V(X))=\sum_Y K(Y)$. Moreover $K(Y)$ factors over the connected components of Y: every polymer in its defining set

belongs to exactly one component. If distinct polymers $X_1, \dots, X_n$ have connected union Y, their incompatibility graph is connected. Otherwise two nonempty groups would partition their union; connectedness of Y would give coincident or adjacent cubes in different groups and hence an edge between them. For each incompatibility-tree edge, choose coincident or face-adjacent cubes in the two polymers and points where their nets meet these cubes. The two points lie in a box of side lengths $M, M, 2M$, so they can be joined by a segment of length at most $\sqrt{6}M < 3M$. Joining near-minimal nets along the tree proves

$$d_M(Y) \le \sum_i d_M(X_i) + 3(n-1).$$

Using $\|\exp V - 1\| \le 2\|V\|$ and $a_2 = a_5 + \kappa_0$ therefore bounds the product by

$$\exp(-a_5 d_M(Y))\big(2H_{\mathrm{cl}} \exp(3a_5)\big)^n \prod_i \exp(-\kappa_0 d_M(X_i)).$$

Replace distinct sets by ordered tuples with factor $1/n!$, and use the counting bound for subpolymers of Y. The result is

$$\|K(Y)\| \le \exp(-a_5 d_M(Y)) x_Y \exp(x_Y), \qquad x_Y = 2H_{\mathrm{cl}} K_0 \exp(3a_5)|Y|_M.$$

By (83),

$$x_Y \le (d_M(Y)+1)/2,$$
$$x_Y \le 4H_{\mathrm{cl}} K_0 c_X \exp(3a_5) \exp(d_M(Y)/2).$$

Since $4\exp(1/2) \le 7$ and $a_5 - 1 = a_4$, we obtain

$$\|K(Y)\|_{(w;\mathsf{w}_W)} \le H_1 \exp(-a_4 d_M(Y)) =: \mu(Y). \tag{84}$$

Weighted Berezin integration costs no norm here: each paired moment at level zero is $\pm 48^{|J|/2}$, which is exactly covered by the weight $(\sqrt{48})^{|J|}$ of its generators. The corresponding moment is ± 1 at positive levels. The photon measure is a probability measure on its compact box. Hence $\|H(Y)\|_w \le \mu(Y)$. These integrals produce even elements, since a nonzero Berezin moment uses an even number of generators. For compatible polymers the fluctuation modes and supported photon coordinates are disjoint. The two product measures therefore factor, with the external old generators left as coefficients. It follows that

$$\int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta) e^{\sum_X V(X)} = \sum_{\{Y_i\} \text{ compatible}} \prod_i H(Y_i). \tag{85}$$

We now bound the infinite connected series. Let

$$S_n(Y) = \sum_{\bigcup_i Y_i = Y} |u^T(Y_1, \dots, Y_n)| \prod_i \mu(Y_i).$$

For $n \ge 2$, apply (77). For every contributing tree the union estimate and $a_4 = a_6 + 2\kappa_0$ give

$$\prod_i \exp(-a_4 d_M(Y_i)) \le \exp(-a_6 d_M(Y)) \exp(3a_6(n-1)) \prod_i \exp(-2\kappa_0 d_M(Y_i)).$$

Drop the prescribed union, retaining $Y_i \subseteq Y$, and root the tree at 1. Write q_j for the number of children at j. A polymer incompatible with Y' contains one of at most $7|Y'|_M$ cubes consisting of its cubes and their face neighbors. Since $|Z|_M \le c_X(d_M(Z)+1) \le \kappa_0(d_M(Z)+1)$ and $a^q \le q! \exp(a)$, the counting estimate gives

$$\sum_{Z \text{ incompatible with } Y'} \exp(-2\kappa_0 d_M(Z)) |Z|_M^q \le 7q! \exp(\kappa_0) K_0 |Y'|_M,$$
$$\sum_{Z \subseteq Y} \exp(-2\kappa_0 d_M(Z)) |Z|_M^q \le q! \exp(\kappa_0) K_0 |Y|_M.$$

Summing from leaves to root thus costs at most $(7\exp(\kappa_0)K_0)^n|Y|_M\prod_j q_j!$. For prescribed children counts, Cayley's degree formula [Pos19, Proposition 4] counts $(n-2)!/((q_1-1)!\prod_{j\geq 2}q_j!)$ trees, where $q_1\geq 1$ and $\sum_j q_j = n-1$. Consequently

$$\sum_{\text{trees}}\prod_j q_j! = (n-2)!\sum_{(q_j)} q_1 \leq (n-1)!\binom{2n-2}{n-1} \leq (n-1)!4^{n-1}.$$

Collecting the factors, also for $n=1$, gives

$$\frac{S_n(Y)}{n!} \leq 7H_1\exp(\kappa_0)K_0|Y|_M\exp(-a_6 d_M(Y))z_S^{n-1}. \tag{86}$$

Since $z_S\leq 1/2$, sum the geometric series and use $|Y|_M\leq c_X\exp(d_M(Y))$ and $a_6-1=a_3$. The bound becomes

$$98c_X^2K_0^2\exp(\kappa_0+3a_5)H_{\mathrm{cl}}\exp(-a_3 d_M(Y)) \leq c_S' H_{\mathrm{cl}}\exp(-a_3 d_M(Y)),$$

because $\kappa_0+3a_5 = 3\kappa-8\kappa_0-6$. Summing only $n>N$ gives the additional factor 2^{-N}. This proves absolute convergence and the claimed estimates.

It remains to justify the logarithm identity after substitution in (76). The even subalgebra with the coefficient norm is a commutative Banach algebra, and the polymer set is finite. The total norm sum of the connected terms is finite by (86). In the exponential of their sum, the norm sum of all products of connected terms is bounded by the exponential of that total norm sum. Regrouping is therefore permitted. Regroup by monomials in the formal variables z_Y. The coefficients are those of (76), namely one for compatible sets and zero for all other monomials. Its substitution $z_Y = H(Y)$ gives (85). Grouping the connected terms by their union gives $\sum_Y E^{\#}(Y)$ and proves (81). □

6.1. **Polynomial approximation at fixed volume.** There are also polynomial approximants to this logarithm which converge uniformly for all families satisfying the preceding bound, at fixed side. Replace $\exp V - 1$ by $p_N(V) = \sum_{1\leq j\leq N} V^j/j!$ in K, H, and stop the connected series at $n=N$; denote the result $E_N^{\#}$. It is a polynomial in integrals of products of $V(X_i)$, $X_i\subseteq Y$, and has the same bound (82). To verify uniform convergence, write $D = |\mathcal{P}_k|$. Since $\|V(X)\|\leq 1$,

$$\|\exp V(X) - 1 - p_N(V(X))\| \leq \frac{\exp(1)}{(N+1)!}.$$

Telescoping each finite product in K gives

$$\|H(Y) - H_N(Y)\| \leq \frac{4^D D\exp(1)}{(N+1)!}.$$

Thus this difference is at most $\delta_N\mu(Y)$, where

$$\delta_N = \frac{4^D D\exp(1)}{(N+1)!\min_{Y\in\mathcal{P}_k}\mu(Y)} \longrightarrow 0.$$

Replacing the n factors in a connected term costs at most $n\delta_N\prod_i\mu(Y_i)$. Summing (86) with this factor, using $\sum_{n\geq 1} n2^{-(n-1)} = 4$, and adding the tail yields

$$\sup_Y \|E^{\#}(Y) - E_N^{\#}(Y)\| \leq 28\delta_N H_1\exp(\kappa_0)K_0(\ell/M)^3 + 2^{-N}c_S' H_{\mathrm{cl}}.$$

This convergence is uniform in the connections and the allowed data. The displayed error depends on the side; no volume-independent approximation is claimed.

7. Symmetries of the local logarithm

Apply proposition 6.1 with $V = F$, $w = h/4$ and denote its output by $E_k^{\#}(m, \mathscr{E})$. Each connected term assigned to Y is a product of integrated activities on subpolymers of Y. Thus ordinary and admissible pair support, local connection dependence, and dependence only on m' and $\mathscr{E}^r(X)$, $X \subseteq Y$, pass to the sum. The uniform bound and the polynomial approximants prove analyticity of its coefficients in $(A, \mathscr{C})$.

Let $\mathbb{G}_k^L$ denote the chamber symmetry group with translations in $LM\mathbb{Z}^3$, so that it preserves every parameter partition. Its actions and the gauge, charge-conjugation and reality actions are those of section 2.2.

Proposition 7.1. The local logarithm satisfies

$$a_{\tilde{g}} E_k^{\#}(\tilde{g}Y; \tilde{g}A, \tilde{g}\mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}), \qquad \tilde{g} \in \mathbb{G}_k^L, \tag{87}$$

$$G_{\lambda+\lambda'} E_k^{\#}(Y; A^1, \mathscr{C}^1) = E_k^{\#}(Y; A, \mathscr{C}), \tag{88}$$

$$\mathrm{Cg}\, E_k^{\#}(Y; -A, -\mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{89}$$

In the gauge identity both pairs belong to the domain and $A^1 = A - d\lambda$, $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on all bonds at distance at most 1 from Y, with real λ, λ'. Here $d\lambda$ denotes the lattice gradient, as in the preceding gauge conventions. For real m and $\mathscr{E} \in K_k^x$ one also has

$$\mathbf{K} E_k^{\#}(Y; -\overline{A}, -\overline{\mathscr{C}}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{90}$$

Every monomial is balanced. ∎

Proof. We first record how transformations pass through the finite operations. Each of the four algebra maps is a continuous ring automorphism, linear for the first three and antilinear for $\mathbf{K}$. If an odd field transforms like the generators it replaces, the two compositions of the field substitution with the automorphism agree on each generator; hence they agree on the algebra. The same observation applies to adding a field. Real-coefficient finite differences (58) then commute with these actions. A lattice symmetry also permutes the parameter cubes and localization summands. It preserves face connectedness, incidence relations and the Ursell coefficient.

The contour functional commutes with the linear actions. For the antilinear action, parametrize $t = r\exp(i\theta)$:

$$\mathcal{C}_r f = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(r\exp(i\theta))}{r\exp(i\theta) - 1}\, \mathrm{d}\theta.$$

Conjugation of coefficients, together with $t \mapsto \bar{t}$ and $\theta \mapsto -\theta$, gives the same integral at the transformed other arguments. The integration over W commutes with every linear or antilinear action which sends $W \mapsto UW$, $\bar{W} \mapsto (U^{-1})^T \bar{W}$. Indeed Wick's rule preserves every paired moment under this change, and all these moments are real. This applies to the lattice and gauge actions, to $\mathbf{K}$ with $U = \varkappa$, and to charge conjugation extended by the identity on W. The last extension will require a separate covariance argument below.

For a lattice symmetry the primitive covariance gives

$$Z(\tilde{g}s')[\tilde{g}\zeta] = \tilde{g} Z(s')[\zeta], \qquad H_k(\tilde{g}s'', \tilde{g}B)\tilde{g} = \tilde{g} H_k(s'', B), \qquad \Gamma_k(\tilde{g}s'', \tilde{g}B)\tilde{g} = \tilde{g}\Gamma_k(s'', B).$$

The barred action is $(S^{-1})^T$ on spinors; $(S^{-1})^T = \pm \mathsf{C}^{-1} S \mathsf{C}$ gives the same covariance for $\bar{H}_k$. Hölder quotients commute with the simultaneous transformation of their endpoints and connection. Thus $J_k^{\#}\Psi$ and $W_k^{\#}$ transform like their ordinary and pair generators. The quadratic primitive satisfies $M_k(\tilde{g}X, \tilde{g}B) = \tilde{g} M_k(X, B)\tilde{g}^{-1}$, so its bilinear form is invariant. The mass form is invariant by the change of chamber site and spin indices, and $R\mathscr{E}$ has the incoming lattice symmetry. The reconstruction substitution commutes with the action by (44). The box measure is invariant under $\zeta \mapsto \tilde{g}\zeta$. Apply these identities to G_1, G_2, G_3, the first localization and substitution, then to the second localization, K, H, and the convergent connected sum. They prove (87).

For the gauge identity set $\Lambda = \lambda + \lambda'$. In a summand assigned to Y, all parameters outside Y vanish. The local row result proves that B^1 may be replaced by $B - d\Lambda$ in every row used in that summand. The latter field is still in the operator domain, since real gauge shifts preserve it. Telescoping the gradient along each path gives

$$U_{B-d\Lambda}(x,x') = U_B(x,x') \exp\big(- ie(\Lambda(c') - \Lambda(c))\big).$$

The ordinary primitive row obeys

$$H_k(s'', B^1)(x,\cdot) = \exp(ie\Lambda(c)) H_k(s'', B)(x,\cdot) \exp(-ie\Lambda^u),$$

when it is nonzero, where Λ^u is the function on the unit indices. The barred row has the inverse phases. In the Hölder row the phase at its second point cancels that from the transport, leaving exactly $\exp(ie\Lambda(c))$, the phase of the first point. The row of Γ_k has the corresponding two-sided gauge covariance. Its composition with $H_k^{\#}$ thus gives the same covariance for $W_k^{\#}$, including the rows which vanish outside Y. The primitive bilinear and the local mass form are gauge invariant, and so is $R\mathscr{E}$ under the separate shifts of its two slots. The reconstruction substitution also commutes with this local gauge action. The photon fields and photon measure are unchanged. The finite operations and integration therefore prove (88).

For reality, the photon primitives are real on real parameter families. Their analytic dependence extends this identity to $\overline{O(\bar{s}')} = O(s')$ on the full polydisk. Since C is real,

$$Z(\bar{s}')[-\zeta] = -\overline{Z(s')[\zeta]}.$$

Transform $s \mapsto \bar{s}$, $t \mapsto \bar{t}$, $\zeta \mapsto -\zeta$ and $(A, \mathscr{C}) \mapsto (-\bar{A}, -\overline{\mathscr{C}})$. The fermion identities are

$$\overline{H_k(\bar{s}'', -\bar{B})} = \varkappa^{-1} H_k(s'', B)\varkappa, \qquad \overline{\Gamma_k(\bar{s}'', -\bar{B})} = \varkappa^{-1}\Gamma_k(s'', B)\varkappa.$$

For the barred field use $\varkappa \mathsf{C} = -\mathsf{C}\varkappa$. The transport relation $\overline{U_{-\bar{B}}(x,x')} = U_B(x,x')$ then gives the same identities for the Hölder rows. The local quadratic primitive obeys the corresponding identity, and $\varkappa^T\varkappa = 1$ preserves the mass form. The number m' is real for the present data, and $R\mathscr{E}$ and the reconstruction substitution have the reality symmetry. The contour argument just given, invariance of the box under sign reversal, and weighted integration prove (90).

Charge conjugation gives the first-localization symmetry by

$$H_k(s'', -B)\mathsf{C} = \mathsf{C}\bar{H}_k(s'', B), \qquad \bar{H}_k(s'', -B)\mathsf{C} = \mathsf{C}H_k(s'', B),$$
$$M_k(X, -B)^T = \mathsf{C}^{-1} M_k(X, B)\mathsf{C}.$$

Anticommuting the two transformed legs verifies the bilinear and mass-form identities. The first-step vertex has this symmetry by its explicit two-half formula. It follows that

$$\mathrm{Cg}\, E_{\le 3}(X; -A, -\mathscr{C}, -\zeta) = E_{\le 3}(X; A, \mathscr{C}, \zeta).$$

Direct covariance of the second substitution is not available, since its unbarred and barred fluctuations use different factors. We instead compare their Wick covariances.

Introduce independent generators $\phi^{\#}$ indexed like $\psi^{\#}$, with their own charge-conjugation action, and let ad add them to the old generators. Put

$$\begin{aligned}\mathcal{V}(X) = {} & \mathrm{ad}\, E_{\le 3}(X) + \mathrm{ad}\, \mathscr{E}^r(X) - \mathscr{E}^r(X) \\ & - \mathbf{1}_{X=\square} m'\big(\mathrm{ad}\langle\bar{\psi}, \psi\rangle_\square - \langle\bar{\psi}, \psi\rangle_\square\big).\end{aligned}$$

It has charge-conjugation symmetry under $(A, \mathscr{C}, \zeta) \mapsto (-A, -\mathscr{C}, -\zeta)$. Let $I_{s'',B}$ fix the old generators and replace $\phi^{\#}$ by $W_k^{\#}(s'', B)$. The second-localization input is exactly $I_{s'',B}\mathcal{V}(X)$, since the polynomial contours in G_4, G_5 give their value at 1 minus their value at 0. For any union U, interpolation restricted to cubes of U, with the disconnected terms removed, gives

$$\sum_{Y \subseteq U} F(Y) = I_{\mathbf{1}_U, B} \sum_{X \subseteq U} \mathcal{V}(X). \tag{91}$$

At $s'' = \mathbf{1}_U$ define

$$\phi_\pm = H_k(\mathbf{1}_U, \pm B)\Gamma_k(\mathbf{1}_U, \pm B)W, \qquad \bar\phi_\pm = \bar H_k(\mathbf{1}_U, \pm B)\bar W, \qquad \phi' = \mathsf{C}\bar\phi_-, \quad \bar\phi' = \mathsf{C}^{-1}\phi_-.$$

The substitution by $(\phi', \bar\phi')$ and their Hölder quotients, denoted I', satisfies $\mathrm{Cg}\, I_{\mathbf{1}_U, -B} = I'\mathrm{Cg}$ on every generator, when Cg fixes W. The only nonzero ordinary pair expectations for the positive substitution are the kernel of

$$\mathcal{G}_U(B) = H_k(\mathbf{1}_U, B)\Gamma_k(\mathbf{1}_U, B)\bar H_k(\mathbf{1}_U, B)^T.$$

The identity moment has kernel $\mathbf{1}_{u=u'}/\nu(u)$, with $\nu = 1/48$ at zero. This precisely cancels one of the two measures in the kernel composition, so the same formula holds at every level. The primitive transpose relation gives

$$\mathcal{G}_U(-B)^T = -\mathsf{C}^{-1}\mathcal{G}_U(B)\mathsf{C}^{-1}.$$

Anticommuting $\bar\phi_-$ and ϕ_- therefore gives

$$\int \mathrm{d}\mu_I(W)\, \phi'(x)\bar\phi'(x')^T = \mathsf{C}[-\mathcal{G}_U(-B)^T(x, x')]\mathsf{C} = \mathcal{G}_U(B)(x, x').$$

Same-bar pair expectations vanish in both substitutions. The pair generators are obtained by the identical linear Hölder maps at $B, -B$, so their pair expectations also agree. Wick's theorem now proves $\int \mathrm{d}\mu_I(W) I'Q = \int \mathrm{d}\mu_I(W) I_{\mathbf{1}_U, B} Q$ for every polynomial Q in the old and $\phi^\#$ generators: each monomial integral is the Pfaffian of its pair expectations, with old generators as coefficients.

Apply this equality to the exponential of $\sum_{X\subseteq U} \mathcal{V}(X)$ and use $\zeta \mapsto -\zeta$. By (91), the integrated subvolume expression $\Xi_U = \int \mathrm{d}\mu_I(W)\mathrm{d}\widehat\mu(\zeta) \exp(\sum_{Y\subseteq U} F(Y))$ has charge-conjugation symmetry. But $\Xi_U = \sum_{Z\subseteq U} H(Z)$, by the finite Mayer expansion. Inverting this finite subset sum gives

$$H(Z) = \sum_{U\subseteq Z} (-1)^{|Z\setminus U|_M} \Xi_U.$$

Thus each $H(Z)$ has the symmetry, and the real coefficients and absolute convergence of the connected series prove (89).

Finally take a constant λ and $\lambda' = 0$ in (88). A monomial with n_u unbarred and n_b barred legs acquires the factor $\exp(ie\lambda(n_u - n_b))$. Invariance for every real λ and independence of the basis monomials force its coefficient to vanish unless $n_u = n_b$. □

8. Scaling and coupling estimates

Let S_L be the field scaling of (30) of [NC]. The nonlinear correction is

$$E_k^*(m, \mathscr{E})(Y; \mathscr{A}, \mathscr{C}) = \sum_{\bar X^L = LY} S_L E_k^\#(m, \mathscr{E})(X; \mathscr{A}_L, \mathscr{C}_L). \tag{92}$$

This formula uses field scaling and summation over fine polymers. Quadratic adjustment enters the incoming activity term in the full step below. The domain inclusions (29) of [NC] ensure $(\mathscr{A}_L, \mathscr{C}_L) \in \widetilde R_k^- \times \mathfrak{X}_k$.

Proposition 8.1. For $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(4)$,

$$E_k^*(m, \mathscr{E}) \in K_{k+1}^{x,\mathbb{C}}, \qquad \|E_k^*(m, \mathscr{E})\|_{k+1}^x \le e^{1/4-6\varepsilon}.$$

For real data its value belongs to K_{k+1}^x. On any affine complex line in the data space its value is norm continuous on the part in the closed ball and norm analytic in its interior. For two data in $\mathrm{Ball}_k^x(1)$,

$$\|E_k^*(m_1, \mathscr{E}_1) - E_k^*(m_2, \mathscr{E}_2)\|_{k+1}^x \le e^{1/4-6\varepsilon}\left(\frac{|m_1 - m_2|}{w_k^{\mathrm{m}}} + \frac{\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x}{w_k^{\mathrm{E}}}\right). \tag{93}$$

The second quotient is absent at $k = 0$. Restriction to $\mathscr{C} = 0$ gives the correction formed without an external slot from $(m, \mathscr{E}|_0)$. ■

Proof. The scaling identity gives

$$\|S_L Q\|_{h_{k+1}} = \|Q\|_{(L^{-1}h_{k+1,1}, L^{-1-\alpha}h_{k+1,2})} \le \|Q\|_{h_k/4},$$

since $h_{k+1,1} = L^{-1/8}h_{k,1}$, $h_{k+1,2} = L^{-1/8+\varepsilon/2}h_{k,2}$ and $L \ge 4$. The polymer sum in (92) uses $Ld_M(Y) \le d_M(X)$ and $X \subseteq LY$. Splitting $a_3 = (a_3 - \kappa_0) + \kappa_0$ therefore yields

$$\begin{aligned}\sum_{\bar{X}^L = LY} \exp(-a_3 d_M(X)) &\le \exp(-L(a_3-\kappa_0)d_M(Y)) \sum_{X \subseteq LY} \exp(-\kappa_0 d_M(X)) \\ &\le c_X K_0 L^3 \exp(-[L(\kappa - 6\kappa_0 - 4) - 1]d_M(Y)) \le c_X K_0 L^3 \exp(-\kappa d_M(Y)).\end{aligned}$$

Together with (82) this gives

$$\|E_k^*\|_{k+1}^x \le c_X K_0 L^3 c_S' H_{\mathrm{cl}} = 504 c_X^5 K_0^5 \exp(3\kappa) L^3 e^{1/4-5\varepsilon} \le e^{1/4-6\varepsilon},$$

using $c_S L^3 e^{\varepsilon} \le 1$.

All generators of a summand lie over X or at a pair admissible for X. Scaling maps the ordinary sites into Y; it maps each of the six paths into the corresponding scaled path and maps $(x, x^{(3)})$ to $(x/L, (x/L)^{(3)})$. These pairs are admissible for Y, including intracluster pairs. A bond meeting X scales to one meeting Y, giving local connection dependence. The scaled-domain inclusion and coefficient analyticity give analytic kernels. The map $(S, a) \mapsto (S, a/L)$ identifies $\mathbb{G}_k^L$ with $\mathbb{G}_{k+1}$ and intertwines its generator actions with S_L. For gauge transformations use $\Lambda_L(c) = L^{1/2}\Lambda(c/L)$ and $(d\Lambda)_L = d(\Lambda_L)$. A gauge relation on the unit neighborhood of Y then holds on the unit neighborhood of every $X \subseteq LY$. Charge conjugation and reality also commute with the real scaling factors. Proposition 7.1 thus gives all the defining symmetries of $K_{k+1}^{x,\mathbb{C}}$, and the real space assertion follows from its reality clause.

For norm analyticity, take data $(m_a + tm_b, \mathscr{E}_a + t\mathscr{E}_b)$ in the closed ball and denote their parameter set by T. It is closed and convex. The finite activity $F_t(X)$ is affine in t by proposition 5.4. The polynomial approximants to $E_k^{\#}$ are therefore polynomials in t. Reblock them by (92); the resulting polynomials $f_N(t)$ converge uniformly on T in the full activity norm, since both finite polymer sets are fixed and the error bound in section 6.1 is uniform. Their values have bounded full norm. If T has more than one point, it contains infinitely many points. The coefficients of each polynomial are finite numerical linear combinations of its values at distinct points of T, by polynomial interpolation. Hence these coefficients have finite full norm, and the uniform limit is norm continuous on T. If T is empty or a singleton there is no continuity or interior analyticity assertion to check.

For a closed disk $|t - t_0| \le r$ in T, take its coefficientwise Cauchy integrals

$$f_j = \frac{1}{2\pi i} \int_{|t-t_0|=r} f(t)(t-t_0)^{-j-1}\,\mathrm{d}t.$$

They have analytic connection kernels, and the support, symmetry and balance conditions pass through this complex linear integral. Moreover $\|f_j\|_{k+1}^x \le e^{1/4-6\varepsilon} r^{-j}$. Coefficientwise Cauchy's formula, with this summable norm bound, gives $f(t) = \sum_{j\ge 0} f_j (t-t_0)^j$ in activity norm for $|t - t_0| < r$. This proves the stated analyticity.

To obtain (93), vary first the mass, keeping $\mathscr{E}_1$ fixed. With $\Delta m = m_1 - m_2 \ne 0$, the line $(m_2 + t\Delta m, \mathscr{E}_1)$ remains in the large ball for $|t| \le r = 3w_k^{\mathrm{m}}/|\Delta m|$. Here $r \ge 3/2$. The contour formula on any circle $1 < r' < r$ gives

$$\|f(1) - f(0)\|_{k+1}^x \le \frac{e^{1/4-6\varepsilon}}{r' - 1}.$$

Let $r' \uparrow r$ and use $r - 1 \ge r/3$. The bound is the first quotient on the right of (93). Vary the activity with mass m_2 fixed and radius $3w_k^{\mathrm{E}}/\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x$ to obtain the second quotient. The zero differences require no contour.

Finally restriction commutes with extraction: $m(\mathscr{E}) = m(\mathscr{E}|_0)$ and $(R\mathscr{E})|_0 = R(\mathscr{E}|_0)$. Every fermion operator uses the sum of its first connection and $\mathscr{C}$; photon operations and measures do not use $\mathscr{C}$. At $\mathscr{C} = 0$ the two constructions therefore have identical finite expressions, connected series and scaled corrections. Analyticity without a slot is understood in the first connection alone. □

The possible Hölder legs can also be read directly. In G_2, G_4 an existing pair generator is changed by addition of a fluctuation field; this creates no new pair index. The mass contours contain no pair generators. The reconstruction substitution and the first-step vertex introduce only pairs $(x, x^{(3)})$. Every later sum, product and integral preserves this set of available indices. Thus the correction has only scaled incoming pair indices and these scaled type-3 indices. At $k = 0$ it has only the latter, of length η_1, since the incoming activity is zero.

9. Identification with the exact density

The three terms in (4) have different origins: the normalized incoming activity, the local determinant difference, and the fluctuation logarithm. We first bound the determinant term, then identify their evaluated sum with the exact block density.

The reblocked determinant term belongs to K^x_{k+1}. Indeed the scaled arguments lie in its inverse domain, and each summand is analytic and reads only fine bonds meeting X. These scale to bonds meeting Y. There are no legs. The lattice symmetry permutes the reblocking summands; local gauge shifts of the two slots become the corresponding real shifts on these fine read bonds, where determinant locality and gauge invariance apply. Evenness of E^d_k gives charge conjugation, and its conjugation identity gives reality. The pointwise bound (82) of [FB] and (83) of [NC] show

$$\|\mathcal{L}\delta E^{\det}_k\|^x_{k+1} \le 8c_{\mathrm{rb}}L^3C_dM^3e_k^{1/4} \le e_k^{1/4-2\varepsilon} \le e_{k+1}^{1/4-2\varepsilon}. \tag{94}$$

Here $L(\kappa - \kappa_0 - 1) \ge \kappa$ controls the remaining exponential weight, and the fixed charge condition is $8c_{\mathrm{rb}}L^3C_dM^3e_k^{2\varepsilon} \le 1$. Restriction to $\mathscr{C} = 0$ gives the same result for $\mathcal{L}E^{\det}_k$.

Proof of theorem A. Use the notation of proposition 3.3, with $A = \mathscr{A}'_L$, $B = A + \mathscr{C}$ and $Z = Z(1)$ the full-strength photon fluctuation. The primitive families satisfy their exact identities at full strength: $H_k(1, B) = H_k(B)$, $\Gamma_k(1, B) = \Gamma_k(B)$, and $\sum_X M_k(X, B) = b_k - D_k(B)$ for $k \ge 1$. The induced fields obey

$$\psi^{0,\#} = H^\#_k(B)\Psi^c, \qquad H^\#_k(B)\Psi^f = \psi^{0,\#} + W^\#_k(1, B).$$

Define three evaluation homomorphisms. Let ev send old generators to $\psi^{0,\#}$ and fix W; let ev_1 send them to $H^\#_k(B)\Psi^f$; for $k \ge 1$, let ev_f additionally send $\Psi, \bar\Psi$ to $\Psi^f, \bar\Psi^f$. At zero take $\mathrm{ev}_f = \mathrm{ev}_1$. These maps exist since their images are odd and have the required antiperiodicity. They commute with the finite coefficient integrals; ev also commutes with Berezin integration, since it fixes W.

On the first contours $\mathcal{C}_r f = f(1) - f(0)$. For G_1, summing the local primitives gives the Dirac difference $-\langle\bar\Psi, (D_k(B + Z) - D_k(B))\Psi\rangle_1$. For G_2, G_3, evaluating $\psi^\# + J^\#_k(1, B, Z)\Psi$ gives $H^\#_k(B + Z)\Psi^f$. The evaluations of their contour differences therefore give

$$\mathrm{ev}_f \sum_X \sum_{i\le 3} G_i(1, X) = \mathrm{Act}^x_k[m', \mathscr{E}^r](A + Z, \mathscr{C}; \Psi^f) - \mathrm{Act}^x_k[m', \mathscr{E}^r](A, \mathscr{C}; \Psi^f). \tag{95}$$

At zero the explicit first vertex gives this same identity with the chamber pairing: $D_0(B) = D_B + \mu$ and the mass form has no connection dependence.

The reconstruction left inverse gives, on every local first expression, $\mathrm{ev}_1 \,\mathrm{sub}_{Y,B} = \mathrm{ev}_f$. This follows on the Ψ generators from $T_k(B)H_k(B) = 1$ and on the old generators from their definition. Adding $W^\#_k(1, B)$ to $\psi^\#$ in a polynomial and then applying ev gives ev_1 applied to that polynomial. Apply these identities to (68). The evaluated $G^W_{\le 3}$ sum is (95). The evaluated polynomial contours G_4, G_5 give

$$I^x_k[m', \mathscr{E}^r](A, \mathscr{C}; \Psi^f) - I^x_k[m', \mathscr{E}^r](A, \mathscr{C}; \Psi^c).$$

Consequently

$$\operatorname{ev}\sum_{Y} F(Y) = V_k^x, \qquad \Xi_k^x = \exp\left\{\sum_X \operatorname{ev} E_k^{\#}(m,\mathscr{E})(X;A,\mathscr{C})\right\}.$$

The second equality is (81) evaluated by ev; its evaluation commutes with the integrals as just noted.

The other remaining factor is the determinant quotient. Equation (78) of [FB] gives

$$\frac{\delta Z_k(B)}{\delta Z_k(0)} = \exp\left\{\sum_X \delta E_k^{\det}(X;A,\mathscr{C}) + \sum_X E_k^{\det}(X,\mathscr{C})\right\}.$$

On inserted fields adjustment changes no value: $\operatorname{ev} R\mathscr{E}(X) = \operatorname{ev}(R\mathscr{E})^{\mathrm{nat}}(X)$. The equality is the telescoping reconstruction identity for a small polymer and the definition for a large one. Finally $\operatorname{ev} = \operatorname{ev}' S_L$, where ev' inserts the outgoing fields. Each fine polymer has exactly one outgoing reblocking polymer. Thus the three activity sums combine into the outgoing $\mathscr{E}^+$, and the two scalar external sums combine into (7). Insert these identities in (42). All exponents are even and commute, so their sums give (8), with exactly its previously derived mass, vacuum and determinant normalization.

Integrate over the outgoing generators and apply the exact marginal (40). This gives (9). The integrand is admissible throughout the photon box: Theorem B of [MB] places $A_k^{\min}(A') + \mathrm{Sq}_k\zeta$ in Dom_k, and its reconstructed connection is an affine, continuous function of ζ. The density coefficients are analytic in this connection on its operator and activity domains, hence continuous and integrable on the compact box. The external scalar factor is fixed during this integral. No condition on the harmonic part of A' enters.

The three terms of (4) belong to K_{k+1}^x by Proposition 6.1 of [NC], (94), and proposition 8.1. Their respective norms are bounded by the three terms in (10). Equation (46) of [NC] bounds the first term with the stronger exponent $3\varepsilon/2$. The mass and vacuum extraction bounds give (11) and (12), since

$$0 \le \delta\varepsilon_k^0 \le 12\exp(-p_{0,k}^2/2) \le e_k^7.$$

At zero all incoming extracted terms vanish. The exact algebra uses all $192n_0$ pairs with the weight $1/48$ and integrates to the $4n_1$-pair outgoing density. There is no removed multiplicity or additional vacuum normalization. The displayed zero-level formulas and bounds follow. □

The restriction to zero external connection commutes with all three terms of (4). The determinant difference there becomes $E_k^d(X,\mathscr{A}) - E_k^d(X,0)$, and the external scalar map preserves $\mathfrak{P}_k(\cdot,0) = 0$. Thus (8) and (9) also give the step without a slot, with incoming activity $\mathscr{E}|_0$.

The displayed norm estimates depend on the running charge and the fixed constants, and on no side or harmonic part. They were proved for any primitive system with the stated hypotheses on the present torus. Identifying the outputs from different systems or different volumes requires the separate comparison arguments; it does not follow from these bounds.

References

[FB] "Chamber fermion locality and local integration". Companion manuscript, version of 20 October 2026. 2026 (cit. on pp. 1–3, 7–12, 14, 17, 20, 31, 32).

[MB] "Maxwell block minimization and Gaussian fluctuations". Companion manuscript, version of 20 October 2026. 2026 (cit. on pp. 2, 7, 11, 12, 17, 32).

[NC] "Relevant extraction and contraction of local interactions". Companion manuscript, version of 20 October 2026. 2026 (cit. on pp. 1–5, 8, 13, 14, 22, 29, 31, 32).

[Pos19] A. Postnikov. *Algebraic Combinatorics*. MIT 18.212, Lecture 23, notes by Andrew Lin, Proposition 4. 2019. URL: https://ocw.mit.edu/courses/18-212-algebraic-combinatorics-spring-2019/f6113618a3e085ba86b799f65934886d_MIT18_212S19_lec23.pdf (cit. on pp. 4, 26).

[QED] “The field-strength Osterwalder–Schrader axioms for QED in three dimensions”. Comprehensive edition, 263 pp., 20 October 2026, before the modular integration revision. 2026 (cit. on pp. 4, 8).